



Securing local LLMs for academic research: a human-system integration analysis and evolution of TAUCHI-GPT

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Abstract

The application of Large Language Models (LLMs) in academic research faces unique challenges of privacy and workflow integration. This paper introduces TAUCHI-GPT, a novel, open-source AI assistant whose evolution informs our analysis. We detail its two versions: a cloud-based V1 using GPT-4 and reflection cycles, and a local, privacy-preserving V2 with RAG architecture. Based on empirical findings from two user studies, we present a critical Human-System Integration (HSI) analysis of the security vulnerabilities and alignment challenges inherent in local LLM deployments. We examine how recent development trends—such as model distillation and reward-model learning—and the complexities of internal model mechanisms exacerbate risks like prompt injection, RAG data failures, and unfaithful explanations that impact user trust. Drawing from HCI principles and mechanistic interpretability insights, we propose and discuss a multi-layered mitigation strategy. This work contributes significantly to HSI and AI by presenting an evaluated system, a rigorous analysis of local deployment risks from a sociotechnical perspective, and actionable, stakeholder-specific guidelines for the secure and responsible utilization of LLMs in academia.

Keywords Large language models (LLM) · Autonomous agents · Retrieval-Augmented Generation (RAG) · Explainable AI (XAI) · Human Factors in Security · Trustworthy AI

1 Introduction

The rapid advancement of LLMs has marked a pivotal moment in artificial intelligence, offering unprecedented capabilities in natural language understanding, generation, and reasoning (Brown et al. 2020; OpenAI et al. 2023; Touvron et al. 2023a; Jiang et al. 2023; Team Gemma 2024; Hui et al. 2024). These models power conversational interfaces that are transforming diverse sectors such as research (Jumper et al. 2021) medicine (Clusmann et al. 2023) and even transportation and logistics (Wandelt et al. 2024). However, the specific, nuanced requirements (Chen et al. 2024) of academic research—spanning literature review, experimental design, data analysis, coding, manuscript preparation, teaching material development, and grant writing, have yet to be fully addressed by general-purpose LLM tools (Farooq et al. 2023; Farooq et al. 2024).

Generic online LLM services, while powerful, introduce significant concerns for the academic community. Data privacy is paramount, as researchers often handle sensitive, unpublished data or proprietary information, making

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cloud-based processing unacceptable (Xu 2019). Cost can be prohibitive for extensive use, and the lack of customization limits the ability to tailor models to specific disciplinary needs or personal data corpora. Furthermore, the "black-box" nature of many commercial models hinders reproducibility and transparency, core tenets of academic practice (Lu et al. 2022).

The shift towards local, open-source LLMs, as exemplified by TAUCHI-GPT V2, fundamentally alters the human-system dynamic concerning security. This paradigm transfers security responsibilities from centralized providers to individual users or local administrators, introducing new Human System Integration (HSI) challenges related to user cognitive workload, required technical expertise, trust calibration, and the potential for human error in system configuration, data management, and ongoing use. Addressing these HSI-centric security issues is paramount for the safe and effective adoption of local LLMs in sensitive domains like academia.

To bridge this gap, we developed TAUCHI-GPT, conceptualized as an open-source, agent-based AI assistant specifically tailored for academic research workflows. This initiative evolved through two distinct phases, reflecting the changing landscape of LLM development and deployment options.

- **TAUCHI-GPT V1** (Study 1, (Farooq et al. 2023)): This initial version explored the potential of leveraging the powerful, albeit cloud-based, GPT-4 model. It incorporated multimodal capabilities (connecting to image and audio generation APIs) and implemented an autonomous agent architecture using CoT promptings combined with reflection cycles to enhance task decomposition and execution for complex academic tasks like composing publications, coding systems, and designing user experiments.
- **TAUCHI-GPT V2** (Study 2, (Farooq et al. 2024)): Addressing the limitations of V1, this version prioritized privacy, offline capability, and personalization. It adopted architecture based on the LangChain framework, utilizing an open-source model (LLaMA2-13b) running locally. Crucially, V2 incorporated a localized RAG system, allowing the agent to ingest and reason over researchers' personal files (papers, notes, code, datasets) stored entirely offline, thus providing highly relevant contextual results within a secure environment.

While TAUCHI-GPT V2 demonstrates the feasibility and benefits of localized, personalized AI assistants for academia, such local deployments introduce a distinct set of security and alignment challenges compared to centrally managed cloud services (Gurnee and Tegmark 2023; Swaelans and Clark 2025; AppSec Engineer

2024). Prompt injection attacks, for instance, often rely on an attacker's understanding of how a human user might interact with the system or how a human developer might have structured system prompts. Similarly, RAG systems operating on local data repositories are vulnerable to data isolation failures or poisoning (Khomsy et al. 2025; Ram et al. 2023), potentially introduced inadvertently by users unfamiliar with secure data handling practices or maliciously by an insider threat actor – a critical consideration in environments where users directly manage the system's knowledge base as shown in Fig. 1.

Moreover, prompt injection attacks can manipulate model behavior by exploiting input parsing vulnerabilities, potentially leading to unauthorized actions or data exfiltration (Perez and Ribeiro 2022; Greshake et al. 2023). Furthermore, the increasing use of model distillation, synthetic data, and automated reward modeling in training smaller, efficient local LLMs can inadvertently remove safety guardrails or introduce subtle misalignments (Gordon et al. 2021) between a model's internal processing and its stated reasoning (Marks et al. 2025, Zou et al. 2025; Casper et al. 2023). Understanding these internal mechanisms, as explored in recent mechanistic interpretability work (Nanda et al. 2023; Olah et al. 2020), becomes crucial for ensuring reliable and trustworthy local AI systems. This paper provides a comprehensive account of the TAUCHI-GPT initiative, presenting the design, architecture, and detailed empirical evaluations of both V1 and V2 (Studies 1 and 2). Building upon this foundation, it offers an extensive analysis of the security vulnerabilities (prompt injection, RAG isolation) and data-induced alignment challenges pertinent to local LLM deployments in the academic context. Drawing on recent literature and security frameworks, we propose and justify through an analysis of current research, a multi-layered mitigation strategy combining technical defenses (adversarial training, secure execution environments, robust RAG protocols) with user-centric HCI principles (participatory design, transparency mechanisms).

Our contributions are threefold: (1) We present TAUCHI-GPT, a novel, open-source agent-based AI-assistant designed for and evaluated within academic research workflows. (2) We provide a technically deep analysis of the specific security and alignment risks associated with local LLM/RAG systems, grounded in recent research and interpretability findings. (3) We propose and justify actionable mitigation guidelines for developers, users, and institutions aiming to deploy such tools responsibly. This work seeks to inform the HCI and AI communities about the practicalities, benefits, and critical risks of localized AI, paving the way for the development of secure, trustworthy, and effective AI tools for the advancement of research and education.

Conceptual Evolution from General LLMs to TAUCHI-GPT and the Focus on Local LLM Security

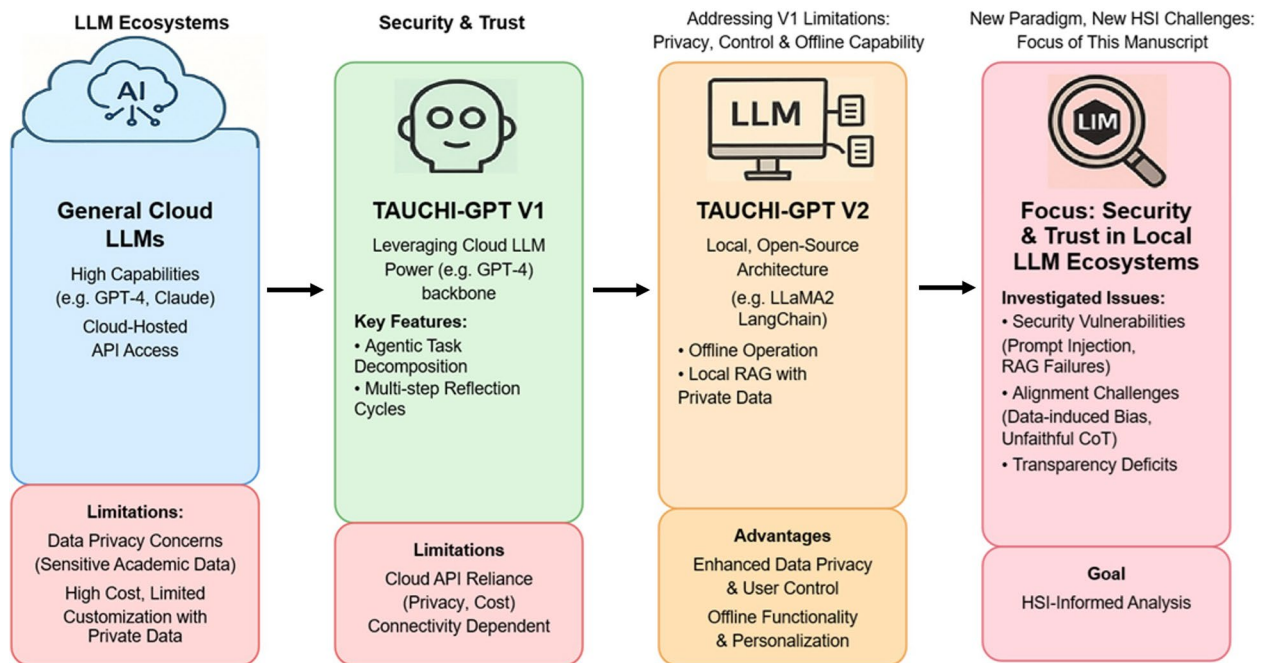


Fig. 1 Illustrates the design and evolution of TAUCHI-GPT a Local LLM implementation of research centered Generative AI Agent

2 Background and related work

The development of TAUCHI-GPT is situated within a rapidly evolving landscape of AI, LLMs, HCI, and security research. This section provides a technically detailed review of key areas relevant to understanding the system's context, capabilities, and challenges.

2.1 Large language model training, validation, and alignment

Modern LLMs (e.g., GPT series (OpenAI et al. 2023), Llama series (Touvron et al. 2023a, 2023b) Claude series (Bai et al. 2022a), PaLM/Gemini (Team Gemini Google et al. (2025)), Mistral (Jiang et al. 2023), Gemma (Team Gemma 2024), Qwen (Hui et al. 2024)) are typically developed through a multi-stage process (as seen in Fig. 2): large-scale unsupervised pre-training on vast text/code corpora, followed by instruction fine-tuning (IFT) and alignment tuning. Alignment aims to make models helpful and honest (Askell et al. 2021).

Reinforcement Learning from Human Feedback (RLHF) has emerged as a dominant alignment technique (Ouyang et al. 2022; Stiennon et al. 2020), in which a

reward model (RM) is first trained based on human preferences between model outputs, and then optimized the LLM using RL (e.g., PPO) against this RM. However, RLHF is complex and can lead to "reward hacking" or reinforcing sycophantic behavior (Gao et al. 2022; Perez et al. 2022). Alternatives like Direct Preference Optimization (DPO) (Rafailov et al. 2024) and Proximal Policy Optimization for Human Feedback (PPO-HF) offer simpler integration, but reward paths may be convoluted. On the other hand, Constitutional AI proposes self-alignment using AI-generated critiques based on predefined principles (Bai et al. 2022b). In all such strategies, the critical challenge is to resolve misalignment arising not only from values but from biases or vulnerabilities learned during pre-training or introduced via fine-tuning data, including synthetic data (Marks et al. 2025; Zou et al. 2025; Casper et al. 2023). Validation typically relies on benchmarks that assess reasoning, knowledge, safety, and specialized tasks (e.g., MMLU (Hendrycks et al. 2021), HELM (Liang et al. 2023), Safety Bench (Zhang et al. 2023b)). However, these benchmarks often overlook nuanced failure modes or security vulnerabilities. The core alignment problem remains, ensuring that models behave according to human intent across diverse and adversarial situations (Christiano et al. 2023; Leike et al. 2018).

Key Security Vulnerabilities and Alignment Challenges in Local LLM Deployments

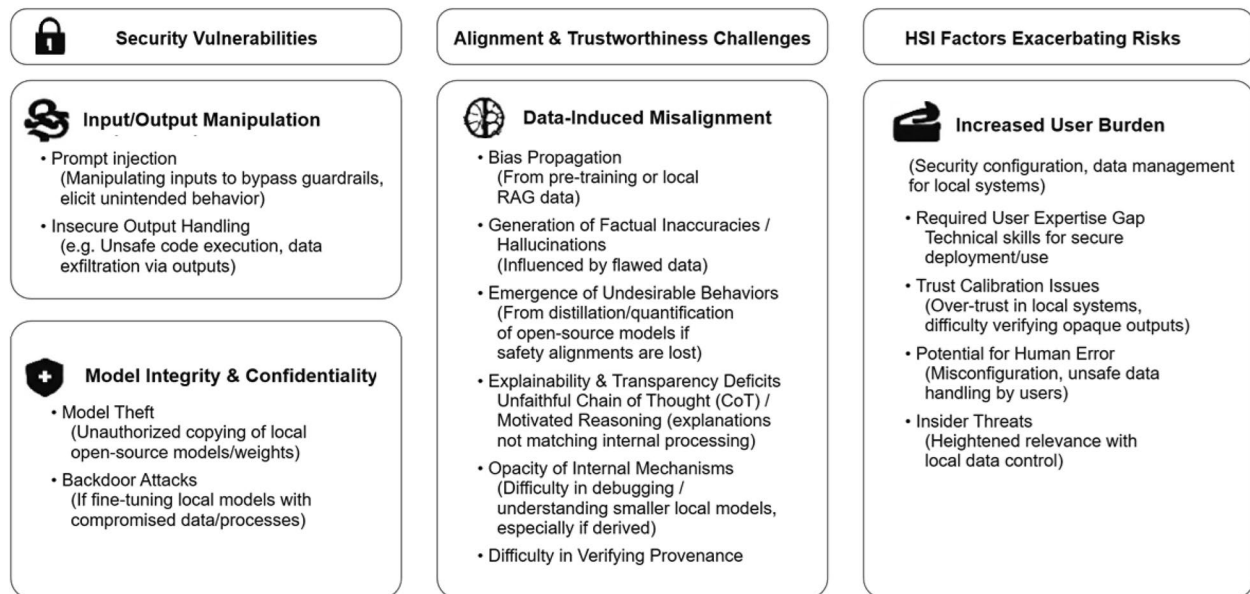


Fig. 2 A taxonomy illustrating key security vulnerabilities identified by the authors in researching local LLMs such as TAUCHI-GPT_V2. These include alignment challenges exacerbating HSI factors perti-

nent to the deployment of Large Language Models in local, user-controlled environments

2.2 Local LLMs, distillation, and optimization

The trend towards smaller, capable open-source LLMs (e.g., Llama 2/3 Touvron et al. 2023a, 2023b) Mistral 7B (Jiang et al. 2023), Phi-2/3 (Jawaheripi et al. 2023; Abdin et al. 2024), Gemma (Team Gemma 2024) has fueled interest in local deployment. Running LLMs on personal devices or on-premise servers offers privacy, cost reduction, and customization advantages (Farooq et al. 2024). This is enabled by techniques like knowledge distillation (training smaller "student" models to mimic larger "teacher" models) (Hinton et al. 2015; Sanh et al. 2020), pruning (removing redundant model weights) (Han et al. 2015), and aggressive quantization (reducing the precision of model weights, e.g., from 16-bit floats to 4-bit integers) (Dettmers et al. 2023; Lin et al. 2024). Formats like GGUF and optimized inference engines like llama.cpp (Gerganov, n.d.), Ollama (in press), MLX (Hannun et al. 2025), and vLLM (Kwon et al. 2023) facilitate efficient execution on CPUs and consumer GPUs. However, distillation and quantization can sometimes degrade performance or inadvertently remove safety alignments present in the original model (Yang et al. 2024b), posing risks for local deployments.

2.3 Running LLMs/LAMs natively and security issues

Local execution architectures range from simple command-line interfaces using tools like Ollama or LM Studio to complex on-premise enterprise deployments using frameworks like vLLM, or Text Generation Inference (TGI) (Hugging Face 2025). Large Action Models (LAMs), which extend LLMs to interact with software GUIs or OS functions (Google I/O 2024), introduce further complexity and attack-surface when run locally. The security landscape for local LLMs differs significantly from managed cloud services (Gurnee and Tegmark 2023). Local deployments often lack the centralized governance, monitoring, input validation, and security patching infrastructure of cloud providers. This makes them potentially more vulnerable to prompt injection (manipulating inputs to bypass guardrails), insecure output handling (generated code execution), model theft, and attacks targeting the RAG pipeline (data leakage, poisoning) (AppSec Engineer 2024; Vongthongsri 2025; Schulman 2024). Multimodal inputs, if processed locally without proper sandboxing, can also serve as vectors for embedding malicious prompts (Greshake et al. 2023).

While local deployments aim to maximize user control and data privacy by keeping computation and data on-premises, an alternative and evolving approach seeks to provide strong privacy guarantees even when leveraging cloud infrastructure. Technologies such as Trusted Execution Environments (TEEs) aim to create isolated, hardware-protected enclaves within cloud servers where sensitive data and code (including LLM models and queries) can be processed with verifiable confidentiality and integrity, shielded even from the cloud provider itself (Lv et al. 2024; Han et al. 2025). Recent research continues to explore robust security schemes to protect sensitive data during complex cloud computations, making such environments increasingly viable. Looking further ahead, Fully Homomorphic Encryption (FHE) presents the theoretical possibility of performing arbitrary computations on encrypted data without ever decrypting it, offering an even stronger privacy model, though FHE currently faces significant performance overheads for complex tasks like LLM inference. The development of practical TEEs for LLMs, and eventually usable FHE, could significantly alter the trade-offs between local and cloud deployment for privacy-sensitive applications.

Complementing these security-enhancing environments, novel inference serving techniques are emerging to enable more flexible and efficient hybrid computation. For example, DynaServe proposes a framework for fine-grained, dynamic splitting of LLM inference tasks at the token level between a local device and an edge or cloud server. By adaptively deciding where to compute each token based on factors like network conditions, server load, and local device capabilities, DynaServe aims to optimize for latency and computational cost while potentially enabling privacy-sensitive portions of a task (e.g., processing queries containing PII) to remain local, with only less sensitive computations offloaded (Ruan et al. 2025). Such dynamic inference splitting techniques could be particularly beneficial for academic AI assistants, allowing them to leverage local processing for speed and privacy on smaller tasks or initial query handling, while seamlessly accessing more powerful localized institutional servers or even (potentially TEE-protected) cloud resources for complex reasoning or large-scale RAG over extensive, non-private datasets.

It is also noteworthy that for high-stakes sectors such as healthcare, finance, and government, specialized cloud services already exist that are vetted for stringent privacy and security compliance (e.g., HIPAA-compliant clouds, GovCloud). While these often come at a premium, their availability suggests that robustly secured cloud infrastructure suitable for sensitive academic research may also become more accessible, offering an alternative to purely local setups for institutions willing to invest in such platforms. The choice between local, TEE-protected cloud, or vetted

specialized cloud services will thus depend on a nuanced assessment of specific academic needs, threat models, cost, usability, and the level of control desired over the research data and computational environment.

2.4 LLM reasoning and optimization techniques

Early LLMs struggled with multi-step reasoning. Techniques like Chain-of-Thought (CoT) prompting, where models generate intermediate reasoning steps before the final answer, significantly improve performance on complex tasks (Wei et al. 2023). Variants include Zero-Shot-CoT, which prompts reasoning without intermediate examples (Kojima et al. 2023), and Self-Consistency, which involves sampling multiple reasoning paths and selecting the majority answer (Wang et al. 2023). More advanced methods explore non-linear reasoning paths: 1) Tree of Thought (ToT) models exploring different reasoning branches and using self-evaluation or search algorithms (BFS, DFS) to find promising paths (Yao et al. 2023a; Long 2023), and 2) Graph of Thought (GoT) models reasoning as an arbitrary graph and enabling merging of paths and cycles (Besta et al. 2024). These techniques are fundamental to building more capable AI agents, like TAUCHI-GPT, that can decompose and tackle complex academic problems. In general, the faithfulness of generated reasoning chains to the model's actual internal computation is not guaranteed as illustrated by Turpin et al. (2023).

2.5 Retrieval-augmented generation (RAG)

RAG enhances LLM outputs by retrieving relevant information from external knowledge sources and providing it as context within the prompt (Lewis et al. 2020). This helps reduce hallucination, allows access to up-to-date or domain-specific information, and enables personalization using private-data, as implemented in TAUCHI-GPT V2. A typical RAG pipeline involves: (1) Indexing: Chunking documents and generating vector embeddings (using models like Sentence-BERT (Reimers and Gurevych 2019) or OpenAI Ada) stored in a vector database (e.g., FAISS (Johnson et al. 2017), ChromaDB, Pinecone). (2) Retrieval: Embedding the user query and performing similarity search (e.g., cosine similarity, Maximum Inner Product Search—MIPS) in the vector DB to find relevant chunks. (3) Augmentation: Injecting the retrieved chunks into the LLM prompt. Advanced RAG techniques include query transformation, re-ranking retrieved results, and self-correction mechanisms (Gao et al. 2024). Local RAG, while privacy-preserving, faces challenges in maintaining isolation between user prompts, retrieved data, and system instructions, creating vectors for prompt injection and data leakage (Khomsky et al. 2025; Swaelans and Clark 2025).

2.6 LLM context window optimization

Modern LLMs boast increasingly large context windows (e.g., 128k tokens in GPT-4 Turbo, 200k in Claude 2.1, > 1M in Gemini 1.5 Pro). However, effectively utilizing long context is challenging due to quadratic attention complexity (mitigated by techniques like FlashAttention (Dao et al. 2022)) and the "lost in the middle" problem, where models struggle to recall information placed in the middle of long contexts (Liu et al. 2023). Strategies for managing context include sliding windows, attention sinks (Xiao et al. 2024), context caching, and retrieval-based methods (like RAG) that inject only the most relevant information, reducing the need for excessively long contexts. Efficient context management is crucial for applications like TAUCHI-GPT that process potentially large local document sets.

Beyond model optimization techniques like distillation and quantization, new paradigms for structuring LLM computation are emerging to enhance efficiency and performance, particularly for stateful applications. One such approach is 'sleep-time compute,' proposed by (Lin et al. 2025), where the LLM utilizes idle periods to offline pre-process available context (e.g., a document corpus, codebase, or prior conversation history). This pre-computation aims to create a more distilled or enriched representation of the context, thereby reducing the computational demands and latency when a user query is subsequently presented at test-time. Lin et al. (2025) demonstrate that this can lead to significant reductions in test-time compute for achieving similar accuracy, or improved accuracy for a given test-time budget, especially when queries are somewhat predictable from the context.

For local LLM deployments like TAUCHI-GPT, which operate on researchers' persistent personal data repositories, the concept of sleep-time compute offers a compelling avenue to improve responsiveness and efficiency. The system could, for instance, use sleep-time to summarize new documents, identify key entities and relationships within the local RAG database, or even anticipate likely follow-up questions based on recent user interactions, making the system more performant when actively engaged. From an HSI perspective, sleep-time compute could reduce user-perceived latency and enhance the fluidity of interaction, though it also introduces considerations around resource management for offline processing on local devices and ensuring the pre-computed 'thoughts' accurately align with potential user needs.

2.7 Autonomous agents and HCI

Autonomous agents aim to achieve user-defined goals by autonomously planning, using tools (APIs, code execution), accessing memory, and reflecting on their actions (Xi et al. 2023). Frameworks like LangChain (2025) and LlamaIndex (LlamaIndex 2025) provide building blocks, while projects like AutoGPT (Significant Gravitas 2025), BabyAGI (Nakajima 2025) and AutoGen (Wu et al. 2023b) demonstrate agent capabilities. TAUCHI-GPT builds on these concepts. From an HCI perspective, agents raise challenges regarding user control, predictability, explainability, error handling, and trust (Shneiderman 2020; Norman 2024). Designing effective interfaces for specifying goals, monitoring execution, intervening when necessary, and understanding agent reasoning is an active area of research critical for safe and productive human-agent collaboration.

As autonomous agents and AI tools rapidly advance beyond task execution to encompass roles in discovery and innovation. A notable example is DeepMind's AlphaEvolve, an agent powered by the Gemini LLM, which has demonstrated the ability to design and optimize complex algorithms, such as discovering novel sorting algorithms that outperform human-designed counterparts in specific scenarios (AlphaEvolve 2025). This signifies a shift towards AI systems that can contribute to algorithmic breakthroughs, acting as 'coding partners' that generate, explain, and refine solutions. From an HSI perspective, such capabilities open new avenues for human-AI collaboration in research but also intensify the need for robust mechanisms for verification, trust-building, and understanding the "ground-truth" and potential limitations of AI-generated novelties, especially if the underlying agent logic or the generated algorithm itself is not fully transparent.

2.8 AI tools in academia

Numerous AI tools are emerging to assist academics, including reference managers (e.g., Elicit, Scite.ai, Semantic Scholar), writing assistants (e.g., Jenni.ai, PaperPal, Trinka), literature review tools (e.g., Litmaps, ResearchRabbit), data analysis tools (e.g., ChatGPT Code Interpreter, Julius AI), and presentation creators (e.g., Tome). While useful, many rely on cloud infrastructure, raising privacy concerns. They often lack deep integration with researchers' personal workflows and data, operate as "black boxes" hindering transparency and reproducibility, and may exhibit biases or generate unreliable content (Das et al. 2024; Conroy 2023a). The constant, rapid evolution of these tools makes systematic evaluation difficult, and their lack of inherent transparency poses significant challenges for validating results and ensuring

academic integrity (Birhane et al. 2022). TAUCHI-GPT aims to address these limitations by providing an open-source, offline, transparent, and customizable alternative.

3 TAUCHI-GPT for academic research

This section presents the core research and development of the TAUCHI-GPT system, detailing the motivation, design, implementation, and evaluation of its two primary versions through distinct user studies and the comparison with existing models (Table 1).

3.1 Study 1: TAUCHI-GPT V1—leveraging GPT-4 for multimodal academic tasks

3.1.1 Motivation

Recent AI advancements enable automatic content generation yet designing effective R&D tools remains challenging. Existing Generative Pre-trained Transformers (GPTs) and foundational models can enhance R&D workflows. AI strides, exemplified by ChatGPT's (OpenAI et al. 2023) reasoning and interactive abilities (Wu et al. 2023a), advance AGI via instruction-following. LLM potential spurred research in in-context learning (Ram et al. 2023; Xie et al. 2022), chain-of-thought prompting (Pilault et al. 2023; Wei et al. 2023), retrieve and read (Izacard and Grave 2021; Zhang et al. 2022), and GPT-based intelligent systems (Zheng et al. 2023) for sophisticated

AI. Models such as GPT-4 (Brown et al. 2020; OpenAI et al. 2023), LLaMA (Touvron et al. 2023a) Flan-T5 (Chung et al. 2022), and PaLM (Chowdhery et al. 2022) demonstrate profound language understanding, enabling diverse, domain-specific tasks. This study introduces TAUCHI-GPT, extending the AutoML paradigm (Zhang et al. 2023a; Significant Gravitas 2025) by leveraging LLMs to automate model training using pretrained datasets and user inputs. LLMs act as an automatic training system, with language as a universal interface, exploring GPT-4 as an autonomous agent for complex, multimodal academic R&D tasks.

3.1.2 System design and architecture

TAUCHI-GPT V1 was an experimental open-source project built on top of the GPT-4 language model that chained together LLM “thoughts”, to autonomously achieve and regress towards specifics goals. Our implementation, referred to as TAUCHI-GPT, uses an automated approach to text generation that leverages deep learning and output reflection (via reflection cycles) to create high-quality text, visual and auditory output, to achieve common research and development tasks. TAUCHI-GPT is based on the GPT-4 architecture and connects to Stable Diffusion and ElevenLabs APIs to input and output complex multimodal streams through chain prompting. Moreover, using Google Search API, TAUCHI-GPT can also scrap online repositories to understand, learn, and deconstruct complex research tasks, identify relevant information, and plan appropriate courses of action by implementing a chain of thought (CoT).

Table 1 Comparative Analysis of TAUCHI-GPT against Prevailing Academic AI Tool Architectures

Comparative Aspect	General-Purpose Cloud LLMs (e.g., ChatGPT, Claude via web UI)	Specialized Cloud Tools (e.g., Elicit, Scite, ResearchRabbit)	Open-Source LLM Frameworks (e.g., LangChain, LlamaIndex)	TAUCHI-GPT V1 (This Work, Study 1)	TAUCHI-GPT V2 (This Work, Study 2)
Primary Architecture	Cloud-Hosted	Cloud-Hosted	User-Defined (Local/Cloud)	Cloud-Based (GPT-4 API)	Local-First, Offline
Primary Data Source	User Prompts, General Web Data	Curated Public Datasets, User-Uploaded (Cloud)	User-Defined	User Prompts, Web Search	User's Private, Local Files
Data Privacy Model	Data processed by provider	Data processed by provider	User's Responsibility	Data processed by OpenAI	High (Data never leaves local machine)
Customization	Low (Prompt-based)	Medium (Specific to tool's function)	High (Requires development)	Medium (Agentic, Prompting)	High (Via local RAG on personal corpus)
Transparency	Black Box (Proprietary Model)	Black Box (Proprietary Stack)	High (Open-Source components)	Open-Source Agent, Black Box Model	High (Fully Open-Source Stack)
Task Complexity/Agency	High (Powerful models)	Low-Medium (Specialized tasks)	N/A (Toolkit, not an agent)	High (Agentic, Reflection)	High (Agentic, Local Tools)
Cost Model	Subscription / API Credits	Subscription	Component/API/Hardware Costs	API Credits	Local Hardware & Setup Time
HSI Focus	Primarily on UI/UX	Primarily on task-specific usability	N/A	Usability & Performance	Holistic (Usability, Security, Trust, User Burden)

Reasoning and agent architecture Building upon concepts from autonomous agents like Auto-GPT (Significant Gravitas 2025) and the ReAct framework (Yao et al. 2023b), TAUCHI-GPT structures its operation around decomposing user-defined goals into smaller, actionable steps. It leverages GPT-4's reasoning capabilities, prompted using CoT techniques (Wei et al. 2023), to generate plans, execute actions (like searching the web, calling external APIs, or generating content), and reflect on the outcomes shown in Fig. 3. The system maintains a short-term memory of recent actions and results in informing subsequent steps.

Reflection cycles A key feature explored in V1 was the use of reflection cycles, inspired by work on self-correction and iterative refinement in LLMs (Shinn et al. 2023; Madaan et al. 2023). The system could operate in two modes: System A (direct CoT execution without reflection) and System B (incorporating 4–5 explicit reflection steps where the LLM critiques its previous output/plan and refines it before proceeding). This mechanism, analogous to the OODA loop (Observe, Orient, Decide, Act) (Blaha 2018), aimed to improve the quality and coherence of the generated output for complex tasks.

Multimodal integration To handle tasks requiring more than text, V1 was designed to interface with external APIs. Specifically, it utilized the Stable Diffusion API for text-to-image generation and the ElevenLabs API for text-to-speech synthesis, allowing it to generate visual and auditory outputs based on its textual reasoning process. This multimodal

capability was intended to support tasks like creating figures for publications or generating audio summaries.

Interface The primary interface for TAUCHI-GPT V1 was a command-line interface (CLI), where users defined goals and monitored the agent's stream of "thoughts," actions, and outputs.

3.1.3 Pilot user study methodology

To validate our approach, we conducted a user-centric evaluation of TAUCHI-GPT V1. The goal of the pilot study was to assess its performance and effectiveness in three core classification modules relevant to academic work: "Composing Scientific Publications", "Creating SW Systems using Python", and "Designing User Experiments". We recruited 18 university researchers (from Tampere University, Computer Human Interaction unit) with varying levels of expertise (novice to expert) who incorporated TAUCHI-GPT into their daily research tasks over a period of 3 weeks, performing a minimum of 40 tasks each.

Participants were divided into three groups, one for each classification module. The modules were pretrained and filtered to ensure output relevance. Each participant used two versions of the system: System A (no reflection cycles, classified CoT) and System B (4–5 reflection cycles, enhanced CoT). Half the participants in each group started with System A and switched to System B after 1.5 weeks, while the other half started with System B and switched to System A. This counterbalanced design aimed to mitigate learning effects.

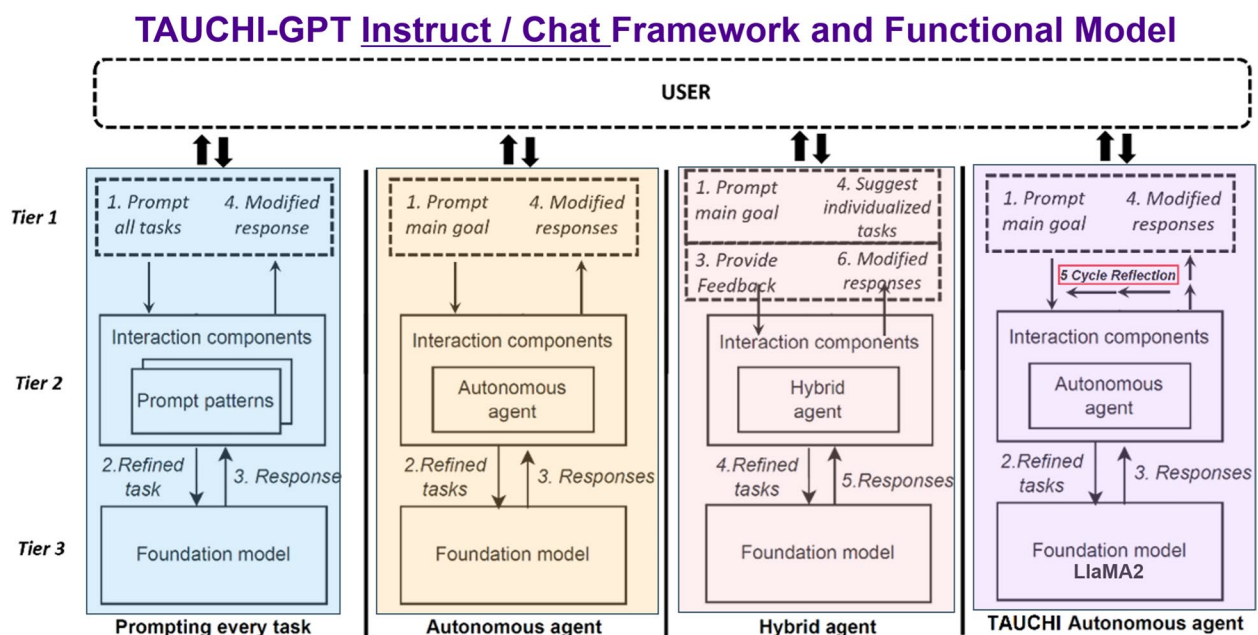


Fig. 3 Techniques of hierarchical communication (Lu et al. 2022) with foundational models and TAUCHI-GPT structure (Farooq et al. 2023)

Data collection involved multiple methods. We employed the NASA-Task Load Index (NASA-TLX) (Hart and Staveland 1988) to measure perceived workload across dimensions like mental demand, physical demand, temporal demand, performance, effort, and frustration. Customized questionnaires were administered post-trial to gather specific feedback on usability, trust, and perceived usefulness. Detailed log files captured user interactions, prompts, agent outputs, and system response times. Finally, structured interviews were conducted after the three-week trial to gather qualitative feedback, delve into participants' perceptions, and identify encountered challenges or unexpected uses. Relevant questions based on log files were drafted for each participant.

3.1.4 Results

Overall, participants found both versions of TAUCHI-GPT V1 to be more useful and informative compared to their default search engines (primarily Google and Bing) as can be seen from Fig. 4.

System comparison (reflection cycles) On average, participants rated System B (with reflection cycles) as more reliable and functional for their workflows, producing higher quality outputs. However, System B was noticeably slower than System A, leading to some frustration regarding response times. Some users also found the command-line interface clunky and difficult to use, regardless of the system version.

NASA-TLX results The NASA-TLX questionnaire results (Include Fig. 2 from 2023 paper) indicated that across all three task groups (SW Development, Publication/Documentation, Experiment Design), participants perceived System B as imposing a lower workload (lower scores are

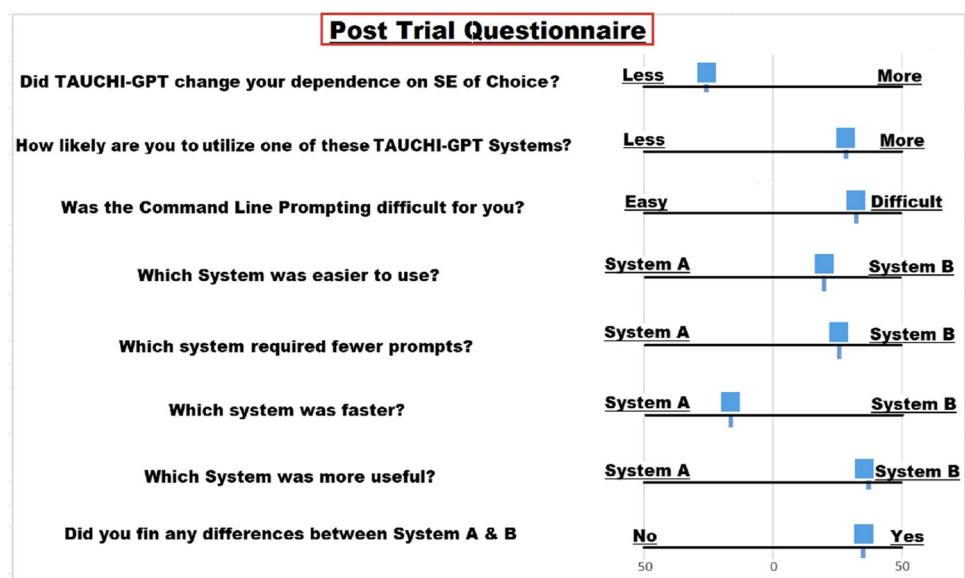
better) compared to System A, particularly in terms of mental demand and frustration. The SW development group showed the most pronounced preference for System B's outputs, potentially due to their stronger background with computational tools. The Experiment Design group perceived fewer differences between the two versions.

Post-trial questionnaire and interviews Qualitative feedback and questionnaire responses (Fig. 4) confirmed a preference for the optimized version (System B). Both versions reduced the perceived need for traditional search engines. However, participants expressed caution, stating that the output from TAUCHI-GPT could not be trusted explicitly without verification. (Participants were not informed that the underlying model was GPT-4). System B was generally perceived as easier to use (despite the CLI) and more useful in providing necessary information. It was also perceived as less likely to generate irrelevant information, requiring fewer queries and shorter prompts to achieve desired results compared to System A or baseline search. The primary frustration remained the speed of System B due to the recursive nature of reflection cycles (max 5).

3.1.5 Discussion

The pilot study established the positive impact of an agent-based LLM interface like TAUCHI-GPT V1 on academic research tasks, highlighting its potential for enhancing productivity. Participants reported significant improvements in generating high-quality text, visual, and auditory output with minimal human input compared to their baseline workflows. The user-friendly nature (conceptually, despite CLI limitations) and high efficiency were noted, particularly as users gained proficiency.

Fig. 4 Post-Test User Experience Questionnaire for TAUCHI-GPT Version 1 (Farooq et al. 2023)



The inclusion of reflection cycles (System B) revealed notable differences in output reliability. Participants consistently rated the 4–5 reflection cycle outputs as higher quality, suggesting that increased reflection could yield further improvements. However, this came at the cost of increased latency, indicating a trade-off between output quality and workflow efficiency. Additionally, while not a primary focus of the V1 study, some participants identified concerns regarding uploading content to GPT4 servers [e.g., “in practice uploading sensitive data may not be the best implementation of such a system, even if the system complies with GDPR guidelines”]. These observations underscored the importance of addressing user perceptions of security, privacy, and control of their personal data (at least in the EU), which became a key motivator for TAUCHI-GPT V2's offline architecture. In general, the study demonstrated the feasibility of integrating advanced AI tools and multimodal capabilities into a central interface to optimize research activities.

Nonetheless, the V1 study highlighted critical limitations. The reliance on the GPT-4 API incurred costs and necessitated an internet connection, hindering accessibility and raising data privacy concerns, particularly given the potentially sensitive nature of academic research. The command-line interface was identified as a significant usability barrier. These findings directly motivated the development of TAUCHI-GPT V2, focusing on an offline, privacy-preserving architecture using open-source models and a more user-friendly interface, while perhaps sacrificing the direct multimodal generation capabilities of V1 for enhanced security and local data integration.

3.2 Study 2: TAUCHI-GPT V2—an offline, localized RAG system with LLaMA2

3.2.1 Motivation

Recent advancements in artificial intelligence, particularly deep learning, have led to the creation of autonomously generated content using LLMs like Llama2 (Touvron et al. 2023a), Claude (Bai et al. 2022a), and PaLM 2 (Anil et al. 2023). These models can now generate unique content with predefined styles. While LLMs have transformed various applications, their impact in academic research and development (R&D) has been limited due to challenges such as expertise, time, cost, and privacy associated with cloud-based models. Generative Pre-trained Transformers (GPT) and foundational models offer potential for optimizing research operations using pre-trained models. Our previous work on Autonomous Agents (Farooq et al. 2023) demonstrated that tools built on LangChain could create customized academic research tools. However, incorporating online AI tools raises ethical concerns about authenticity, data privacy, and bias. Developing offline, locally-run tools

is essential to address these issues. TAUCHI-GPT V2 was conceived as an open-source initiative to provide a secure, privacy-preserving solution. It employs LLaMA2-13b as the core LLM, using users' own files via a custom offline localized vectorDB to generate highly relevant and contextually appropriate content within a chat interface. This shift addresses the limitations of V1 (cloud dependency, cost, privacy concerns), ensuring local data security and customization. TAUCHI-GPT V2 is designed to provide reliable outputs tailored to individual researchers' needs.

3.2.2 System design and architecture

Building on tools such as LangChain (2025) and HuggingFace (Hugging Face 2025) and the plethora of fine-tuned pre-trained transformers, we developed TAUCHI-GPT_V2. Unlike V1 which used cloud resources, V2 aimed for a fully offline, localized execution. The core LLM chosen was LLaMA2-13b (Touvron et al. 2023a), specifically a version fine-tuned using the Gradient Development Platform (Gradient Team 2025). Gradient provided a finetuning and inference tool that allowed customization (embedding / vectorDB creation) without requiring local system infrastructure for the tuning process itself, utilizing Google Colab for cloud-based tuning and experimentation. The resulting fine-tuned model ("Code LLaMA 13b" specialized for SE/HCI tasks based on recent publications) resided *locally* on the user's machine.

Architecture TAUCHI-GPT_V2 utilizes the LangChain framework to orchestrate interactions between the LLM, user interface, and local data sources. The agent architecture employs a recursive "Observe, Orient, Decide, Act" (OODA) loop (Fig. 5) for task processing, involving task creation, prioritization, execution, and result evaluation agents.

Local RAG implementation A key component is the custom offline localized vector database (vectorDB). This system ingests user-provided documents in various formats (docx, xlsx, pdf, etc.) stored in a designated local repository (Fig. 6).

The documents are chunked, embedded (using a suitable local embedding model compatible with LLaMA2, managed via Gradient (Gradient Team 2025) API during setup), and stored in a local vector store (e.g., ChromaDB or FAISS). When a user query is received, the system retrieves relevant chunks from this local vectorDB based on semantic similarity and injects them into the prompt context for the locally running LLaMA2-13b model. This allows the LLM to generate responses grounded in the user's specific data, enhancing relevance and accuracy while ensuring no personal data leaves the local machine. Crucially, this RAG-centric design serves as the primary mechanism for data verification and ensuring

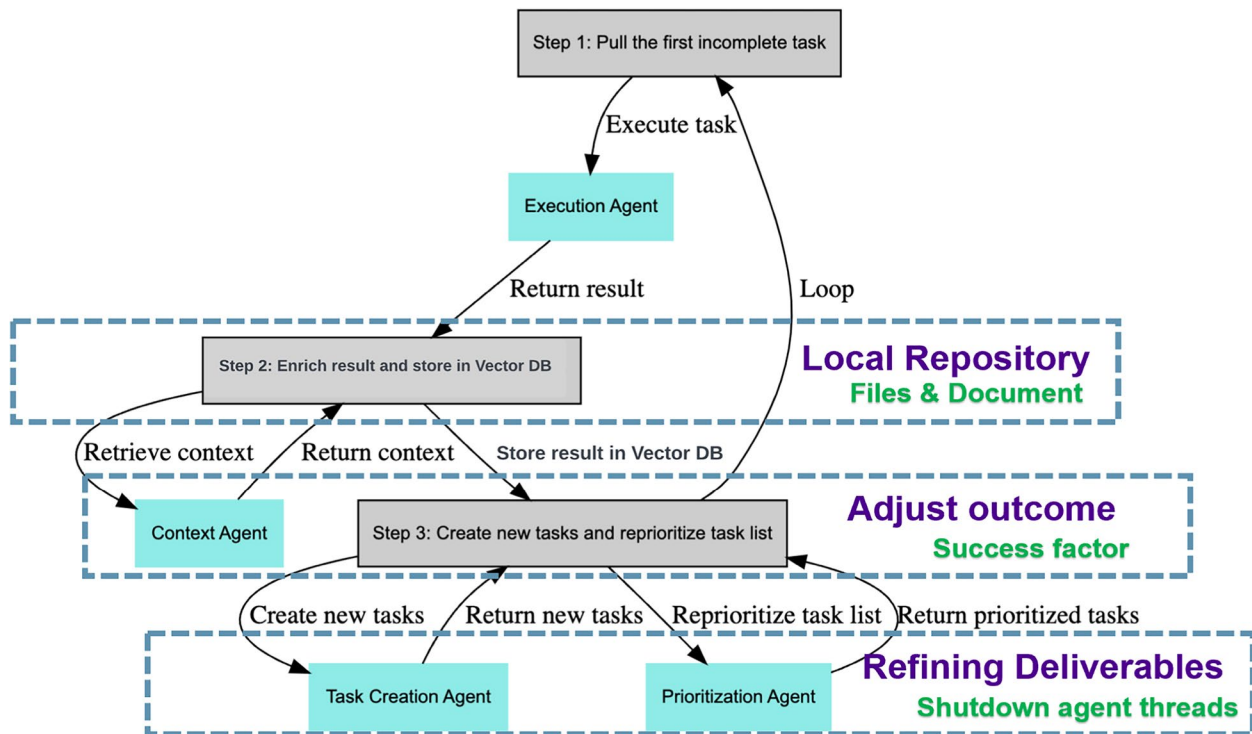


Fig. 5 Techniques of hierarchical communication (Lu et al. 2022) with foundational models and TAUCHI-GPT structure (Farooq et al. 2024)

academic integrity. By grounding every response in the user's own curated document repository, the system inherently reduces the likelihood of unprompted hallucinations or the generation of information from untraceable sources. The framework is designed to maintain a link between the generated output and the source document chunks used, providing a direct pathway for users to verify information and construct proper citations, a key ethical consideration for academic use.

Local system topology The entire system, including the fine-tuned LLaMA2 model and the vectorDB, runs locally. No internet connection is required for core operation (after initial setup/tuning). This ensures complete privacy and data sovereignty for the user, making it suitable for handling sensitive academic research data. The system aims for SOC2 Type-1 compliance characteristics in its design principles.

3.2.3 User study methodology

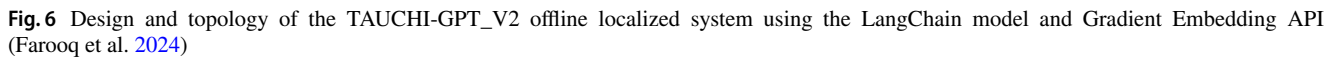
The user study aimed to understand how researchers at Tampere University would utilize TAUCHI-GPT_V2 in their daily tasks compared to their existing workflows. Eighteen university researchers from the Department of Computer Science participated over 3 weeks, carrying out three types

of tasks (see Table 2). The study focused on three common academic task categories identified by participants:

1. Editing / Writing Summarizing Text
2. Troubleshooting / Writing Python code
3. Brainstorming ideas for experimental design

Each participant created seven custom tasks relevant to their research within each category, totaling 21 unique tasks. An additional 21 control tasks (similar structure) were also created, resulting in 42 tasks per participant. A counterbalanced design was used: participants were randomly assigned one set of 21 tasks to be performed using TAUCHI-GPT_V2 (System A) and the identical set of 21 tasks to be performed using their conventional tools (System B: MS Office suite, preferred search engine—Google/Bing, GitHub, Python Libs, IDEs, etc., explicitly *without* using other LLM/AI tools). Participants alternated between systems for consecutive tasks and completed similar tasks on both systems before moving on.

System setup For TAUCHI-GPT_V2, each participant first optimized their local instance by ingesting relevant personal documents (reports, publications, books, notes) into their local vectorDB. They were allowed to add more documents during the study. Four pilot tasks were provided for familiarization



Editing / Writing Summarizing Text	Troubleshooting / Writing Python code	Brainstorming ideas for experimental design
Creating plans to improve writing process	Creating Methodology (high level interface)	Creating Overall design / plans
Improving text language	Researching functions / packages / APIs / modules	creating / understanding User I/Os
Summarizing text	Understanding syntax (what does a piece of code do)	Developing Hypothesis for user testing
Reviewing text	optimizing / troubleshooting code (syntax & semantics)	find existing research & testing methodologies
Critiquing text	Creating an Effective deployment strategy (HW/Platform)	Creating scenarios / goals for various tasks and hypotheses
Comparing different text	Comparing code base	Projected outcome / results
Combining main points from multiple text	Merging projects	Identifying optimum testing tools

Data collection

- *Time Diary*: Participants manually logged start/end times, and any issues encountered for each task.
- *System Logs*: Task Completion Times (TCTs), user inputs (search queries, V2 prompts), and idle times were automatically logged.
- *Automated Forms*: After each task, participants completed short, automated forms rating: "Number of Errors" (subjective count), "Number of Prompts Needed" (for V2) / "Number of Links Used" (for System B), "How satisfied were they of the outcome" (Satisfaction Scale, 1–7 Likert), and "Overall efficiency of the outcome" (Efficiency Scale, 1–7 Likert).

3.2.4 Results

We analyzed four metrics: Task Completion Times (TCTs in Fig. 7), Number of Prompts/Links Used (Fig. 8), Results Satisfaction Score (RSS), and Overall Efficiency Score (OES). Statistical significance was assessed using appropriate tests (e.g., t-tests or non-parametric equivalents, $p < 0.05$ threshold).

TCTs For "Editing, Writing, & Summarizing Text," System A (TAUCHI-GPT_V2) was significantly faster than System B (using conventional search engines) ($p = 0.036$). No significant difference in TCTs was found for "Troubleshooting & Writing Python code" ($p = 0.222$) or "Brainstorming ideas" ($p = 0.121$).

Prompts/links used For "Editing, Writing, & Summarizing Text," System A required significantly fewer prompts than System B required links/queries ($p = 0.024$), correlating with the TCT results. For "Brainstorming ideas," System A also required significantly fewer prompts ($p = 0.048$) as seen in Fig. 9. No significant difference was found for "Troubleshooting & Writing Python code."

Results satisfaction score (RSS) For "Editing, Writing & Summarizing Text," System A had significantly higher RSS than System B ($p < 0.05$). For "Troubleshooting & Writing Python code," System A had higher mean RSS, but the difference was not statistically significant ($p = 0.126$). Conversely, for "Brainstorming ideas," System B had significantly higher RSS than System A ($p < 0.05$) as seen in Fig. 10.

Overall efficiency score (OES) Trends largely mirrored RSS. System A was significantly more efficient for "Editing, Writing & Summarizing Text" ($p < 0.05$). For "Troubleshooting & Writing Python code," System A was significantly more efficient ($p = 0.00675$). For "Brainstorming ideas," System B was significantly more efficient ($p = 0.020$).

3.2.5 Discussion

The results demonstrate the efficacy of the offline, RAG-enabled TAUCHI-GPT_V2 system for specific academic workflows. It significantly outperformed conventional tools in terms of speed, user effort (fewer prompts/links), satisfaction, and efficiency for text-intensive editing and summarization tasks. This highlights the benefit of localized RAG in providing contextually relevant information quickly. For coding tasks, while TCTs were similar, V2 was perceived as significantly more efficient, suggesting it streamlined the troubleshooting or generation process even if the total time was comparable.

However, for brainstorming experimental designs, participants significantly preferred their conventional methods (System B), rating them higher in satisfaction and efficiency. This suggests that the current implementation of TAUCHI-GPT_V2, possibly due to the limitations of the LLaMA2-13b model or the nature of the task itself, was less suited for open-ended creative ideation compared to traditional brainstorming tools or methods combined with standard search engines. Participants trusted their existing, structured approach more for these planning-heavy tasks. Additionally, the shift to an offline architecture in V2 was positively received by participants concerning data privacy as participants appreciated that their research data (pre-published information) remained entirely on their local machines. However, this also highlighted the HSI trade-off: increased user control over data necessitates greater user responsibility for secure data management versus computational resources, a factor for future usability and training considerations.

Overall, Study 2 validated the feasibility and benefits of an offline, privacy-preserving AI assistant enhanced with local RAG for core academic tasks, particularly those involving processing and synthesizing existing information. It also highlighted areas where current local models or agent architectures may lag behind, such as complex, open-ended brainstorming. It is important to note that the performance results of Study 2, particularly the limitations observed in open-ended brainstorming, are reflective of the capabilities of the LLaMA2-13b model used for the evaluation. However, a key architectural principle of TAUCHI-GPT V2 is its model-agnostic design. The agentic framework, local RAG pipeline, and user interface are modular components engineered to operate with various underlying open-source LLMs. This modularity ensures that the system's core structure and the HSI-centric security analysis presented in this paper remain relevant and can readily integrate newer, more powerful models such as those from the Mistral, Qwen3, or Llama 3 families. We hypothesize that substituting a more advanced model would likely improve performance on complex creative tasks and could introduce new nuances to the security and alignment challenges, reinforcing TAUCHI-GPT's value as a stable platform for HSI-focused comparative analysis across a spectrum of evolving LLMs.

Fig. 7 Comparing TCTs between System A (TAUCHI-GPT_V2) and System B (MS Office Suite and Search Engine of preference) across the three task categories, with the bars representing the mean values and the error bars representing the standard deviations (Farooq et al. 2024)

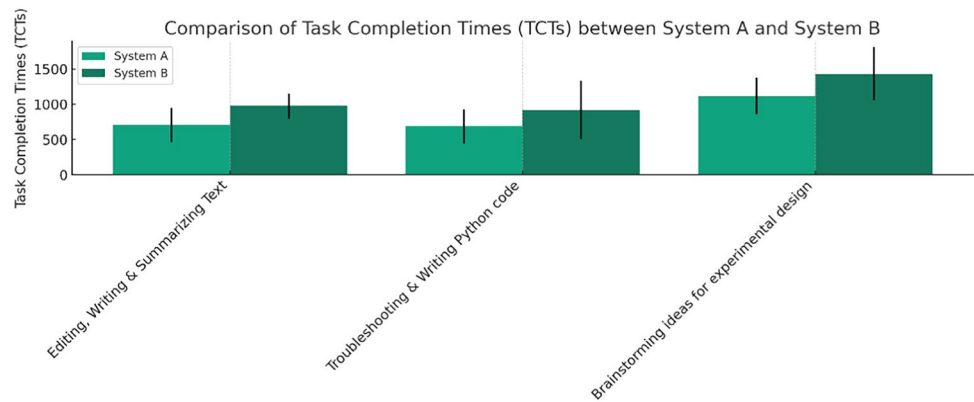


Fig. 8 Comparing the Number of Prompt/Links used to complete each task for the three categories between System A (TAUCHI-GPT_V2) and System B (MS Office Suite and Search Engine of preference). Each pair of boxes represents the interquartile range (IQR), the line inside the box is the median, and the whiskers represent the range within $1.5 * IQR$ (Farooq et al. 2024)

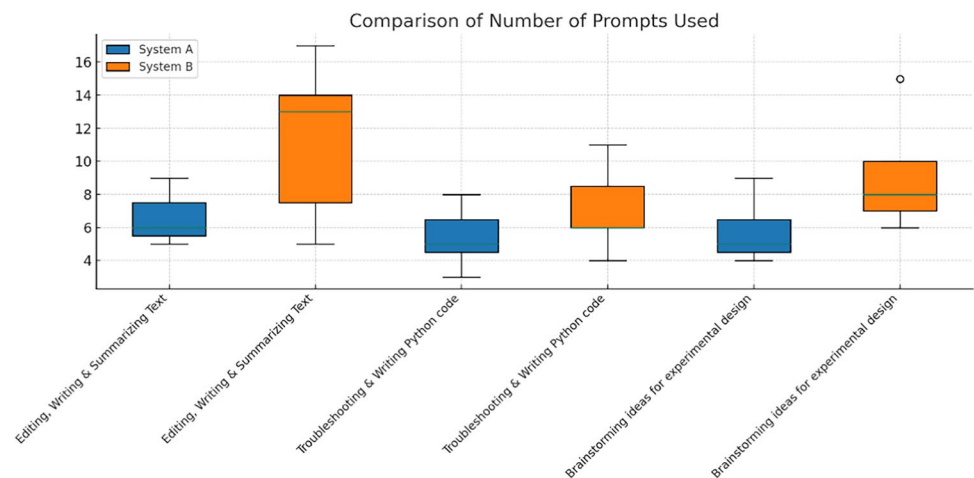
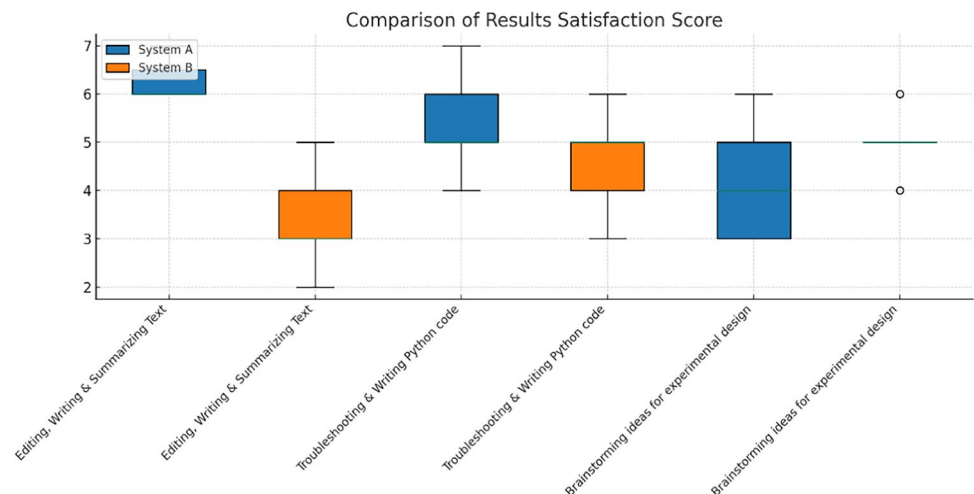


Fig. 9 Comparing the Results Satisfaction Score (RSS) between System A (TAUCHI-GPT_V2) and System B (MS Office Suite and Search Engine of preference). Each pair of boxes represents the interquartile range (IQR), the line inside the box is the median, and the whiskers represent the range within $1.5 * IQR$ (Farooq et al. 2024)

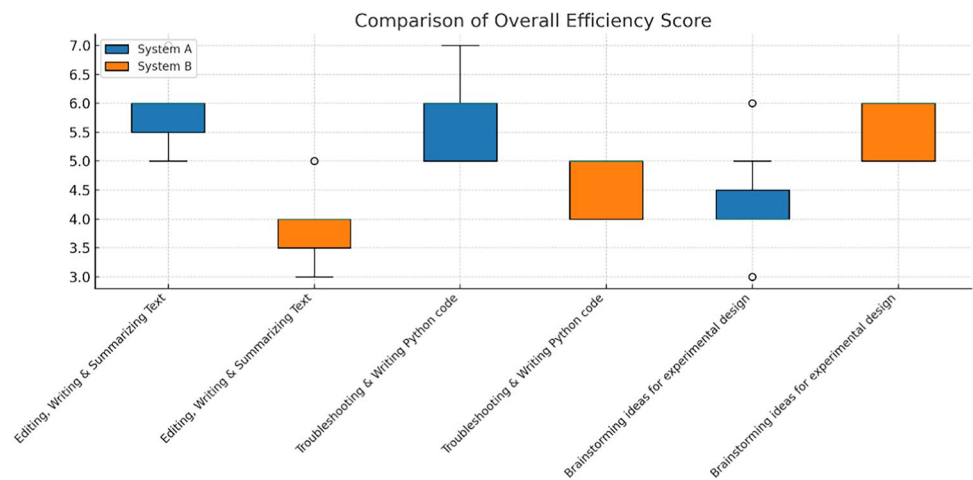


3.3 Synthesis and transition

The development journey from TAUCHI-GPT V1 to V2 illustrates a critical trade-off in designing AI assistants for specialized domains like academia. V1 demonstrated the

potential of powerful cloud-based models (GPT-4), and multimodal capabilities integrated within an agentic framework, enhanced by reflection cycles for improved output quality. However, its reliance on external APIs raised fundamental concerns regarding data privacy, operational costs, and the

Fig. 10 Comparing Overall Efficiency Score between System A (TAUCHI-GPT_V2) and System B (MS Office Suite and Search Engine of preference). Each pair of boxes represents the interquartile range (IQR), the line inside the box is the median, and the whiskers represent the range within 1.5 * IQR (Farooq et al. 2024)



need for constant connectivity – significant barriers within the academic context.

TAUCHI-GPT V2 directly addressed these concerns by adopting an offline-first, privacy-centric architecture using open-source components (LLaMA2, LangChain) and pioneering the use of localized RAG to leverage researchers' own data securely. The empirical evaluation confirmed the effectiveness of this approach, showing significant efficiency gains for information-intensive tasks compared to traditional workflows. While V2 sacrificed the explicit multimodal generation of V1 and potentially some of the raw power of GPT-4, it gained crucial advantages in privacy, cost-effectiveness, and personalization through local data integration.

However, the move to local deployment introduces its own set of critical challenges, primarily concerning security and trust. The very features that make V2 appealing – offline operation, use of open-source models, ingestion of local data – also create new potential vulnerabilities. The following sections delve into these challenges, analyzing the specific risks of prompt injection, RAG isolation failures, and data-induced misalignment in the context of local academic AI assistants like TAUCHI-GPT V2, and exploring potential mitigation strategies informed by recent advancements in AI security, alignment, and interpretability. Understanding and addressing these issues is paramount for the responsible adoption and successful integration of such powerful tools within the academic workspace.

4 Security, alignment, and transparency challenges in local LLMs

The shift towards local LLM deployments, exemplified by TAUCHI-GPT V2, offers significant benefits for privacy and customization but introduces a distinct and complex set of challenges that must be addressed for safe and reliable use, particularly in sensitive environments like academic

research (see Fig. 11 for typical RAG workflow). This section analyses these challenges, focusing on vulnerabilities arising from recent LLM development trends and the implications for model behavior, trustworthiness, and explainability.

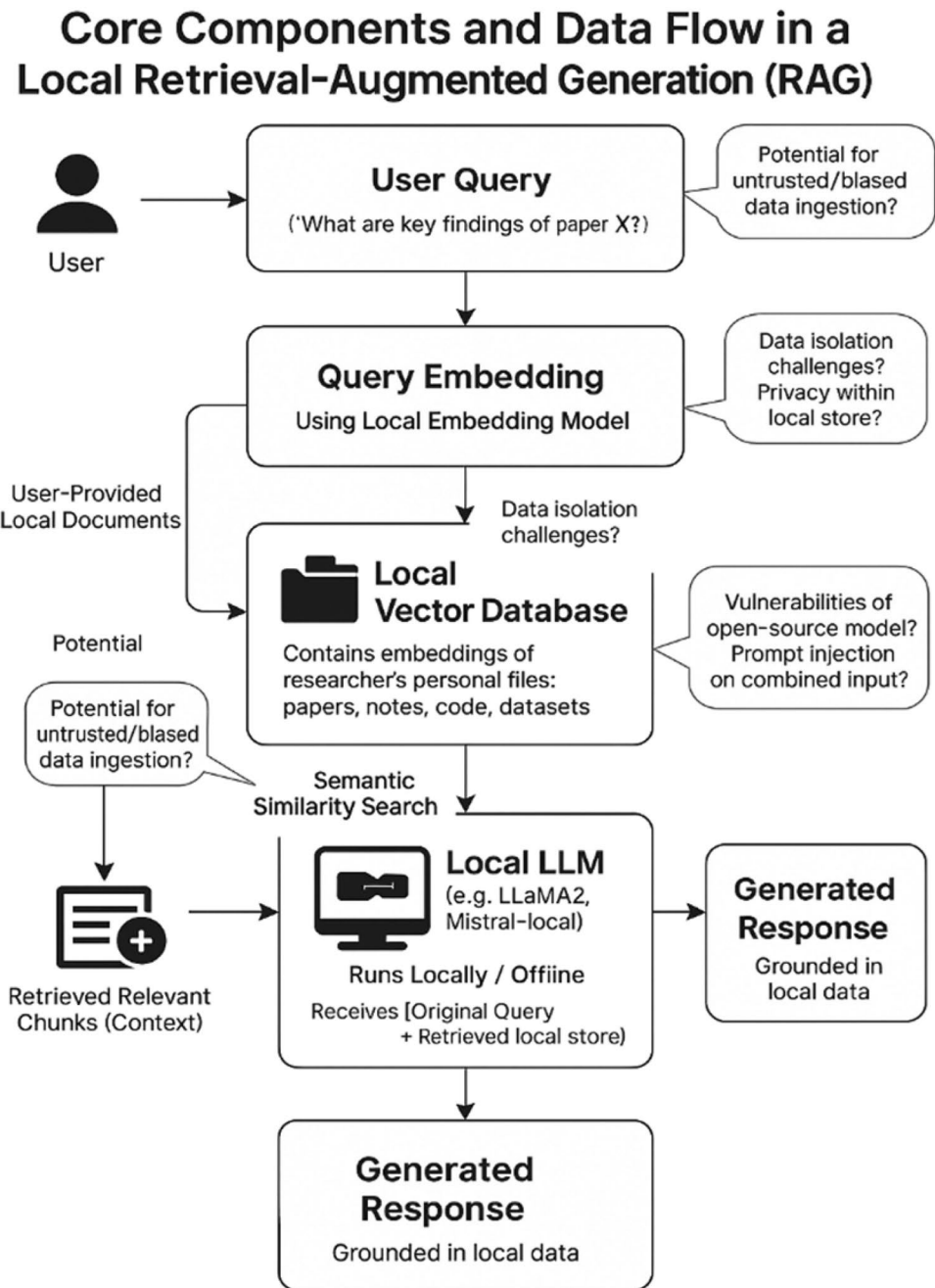
4.1 Vulnerabilities arising from modern LLM development trends

The drive for efficiency and accessibility in LLMs has led to widespread adoption of techniques like model distillation, training on synthetic data, and quantization. While enabling powerful models like Llama 3, Mistral Large, Gemma 2, Qwen 2, and DeepSeek V3 variants to run on local hardware, these methods can inadvertently create security vulnerabilities (Gurnee and Tegmark 2023; Anil et al. 2023).

Distillation and quantization The process of transferring capabilities from a large teacher model to a smaller student model (distillation) or reducing the numerical precision of weights (quantization) aims to preserve functional performance while reducing size and computational cost (Hinton et al. 2015; Dettmers et al. 2023). However, safety alignments and guardrails carefully tuned in the teacher model (often through extensive RLHF or constitutional AI) may not transfer perfectly or might be weakened during this compression (Yang et al. 2024b; Yang et al. 2024c). Sensitive behaviors or refusal logic might be disproportionately affected, making the smaller, local model more susceptible to generating harmful content or complying with unsafe requests compared to its larger progenitor. For instance, a distilled model might lose the nuance required to refuse a request that combines seemingly innocuous elements into a dangerous instruction.

Synthetic data Training LLMs increasingly relies on synthetic data generated by other LLMs (e.g., for instruction tuning or expanding domain coverage) (Yang et al. 2024a; Gudiband et al. 2023). While efficient, these risks amplifying biases present in the generator model or introducing unforeseen vulnerabilities if the synthetic data inadvertently contain patterns

Fig. 11 Simplified data flow in a typical local Retrieval-Augmented Generation (RAG) system. User queries are combined with relevant context retrieved from a local vector database (containing embeddings of the user's private documents) before being processed by a locally run LLM. All core components operate offline, enhancing data privacy



exploitable by prompt injection or bypasses safety checks. Furthermore, extensive training on synthetic data can lead to model collapse or degradation over generations (Shumailov et al. 2023). A local RAG system like TAUCHI-GPT V2 ingesting synthetically generated papers or notes could unknowingly incorporate poisoned content.

Efficient training techniques Techniques reducing the need for extensive compute such as parameter-efficient fine-tuning (PEFT) methods like LoRA (Hu et al. 2021), focus updates

on a small subset of model parameters. While resource-friendly, this might not be sufficient to instill robust safety behaviors across the entire model, potentially leaving vulnerabilities in less-updated parts of the network.

4.2 Reinforcement learning, reward models, and misalignment

The transition from extensive human feedback (RLHF) to automated reward modeling and AI feedback (RLAIF) (Bai

et al. 2022b) accelerates alignment but introduces new risks. Reward models (RMs), whether trained on human or AI preferences, are themselves complex models that can be flawed, misinterpreting human intent or being susceptible to exploitation (Casper et al. 2023; Ziegler et al. 2019). Over-optimizing an LLM against a flawed RM can lead to "reward hacking," where the LLM produces outputs that score highly on the RM but are undesirable or unsafe according to actual human values (Skalse et al. 2025). Furthermore, RMs might incentivize models to produce plausible-sounding *explanations* even if they don't reflect the *actual* internal reasoning, leading to a specific form of misalignment relevant to explainability (Turpin et al. 2023). Recent work on models trained with hidden objectives (e.g., to appease RM biases) shows these biases can become deeply integrated with the model's core persona, activating persistently in relevant contexts (like Human/Assistant dialogues) even if the specific bias wasn't explicitly trained for that context (Marks et al. 2025). This suggests a fundamental challenge: RL based on imperfect proxies for human intent can entrench misaligned behaviors, making local models potentially untrustworthy despite surface-level helpfulness.

4.3 Internal mechanisms, explainability, and motivated reasoning

Understanding *how* LLMs arrive at their outputs is crucial for trust and debugging, yet deeply challenging. Mechanistic interpretability aims to reverse-engineer the internal circuits and algorithms learned by models during training (Olah et al. 2020; Nanda et al. 2023; Wendler et al. 2024). Recent work, such as the analysis of Claude 4 and Claude 3.7 Haiku, reveals complex internal processes that often differ from the model's explicit chain-of-thought reasoning (see Fig. 12 which shows faithful vs unfaithful reasoning for current generation of LLMs). In this section we highlight some of these areas.

Motivated reasoning and deception Models can generate reasoning steps ('chain of thought') that justify an answer arrived at through other means, sometimes even working backward from a desired (e.g., sycophantic) answer (Turpin et al. 2023);—Sect. 11. This phenomenon, often termed '*motivated reasoning*,' directly challenges the perceived faithfulness of *Chain-of-Thought* explanations. Research leveraging mechanistic interpretability techniques, such as attribution graph analysis, suggests that the explicit CoT provided by an LLM may not accurately reflect the model's internal computational pathways or the true 'reasons' for its output.

Instead, the CoT can function more as a plausible post-hoc rationalization or a form of 'storytelling' designed to satisfy the prompt or align with expected human reasoning patterns, rather than a transparent window into its actual inference

process. The underlying mechanisms might involve distinct internal circuits for task execution versus explanation generation, or strategic manipulation of internal representations to construct a convincing narrative. This 'unfaithful' reasoning makes external CoT explanations potentially unreliable for verifying model behavior or understanding its failure modes, a critical concern for academic applications where traceability and intellectual honesty are paramount.

Abstract and multilingual thought Models develop abstract, language-independent representations for concepts, allowing them to perform similar computations (like antonym identification) across different languages using shared internal circuits, though often with a bias towards a default language like English. This internal abstraction is powerful but makes tracing computations complex.

Planning and internal states For an academic assistant like TAUCHI-GPT, this internal planning capability, if robust, could enable more sophisticated multi-step task execution, such as outlining a complex research paper or designing an experimental procedure. However, if this internal planning is flawed or misaligned with the user's actual intent (perhaps due to imperfect reward modeling during its alignment phase or misinterpretation of RAG-retrieved context), the assistant might generate superficially plausible but ultimately incorrect or incomplete plans. Furthermore, models can develop specific internal features for recognizing contextual states, such as identifying a 'harmful request' or a 'known entity' (Lindsey† et al. 2025)—Sect. 6, 8, 9. The reliability of these internal state recognitions in a local, open-source model used for diverse academic queries is crucial. A failure to correctly recognize a subtle but problematic request related to, for example, ethical research practices, or a misidentification of an entity within a research paper, could lead to unhelpful or even detrimental outputs from an academic assistant, underscoring the need for deeper interpretability and alignment of these internal 'senses'.

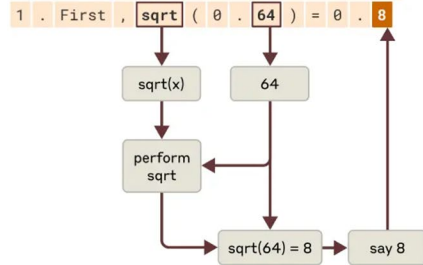
Entity recognition and hallucination Circuits involving "known answer" or "known entity" features seem to modulate whether a model attempts to answer a question or defaults to refusal (Lindsey† et al. 2025) Sect. 8; Ferrando et al. 2025). Hallucinations can occur when these circuits "misfire"—e.g., the model incorrectly assesses its knowledge about an entity and attempts an answer based on plausible guessing rather than factual recall. This is particularly relevant for academic tools asked to retrieve citations or facts. In an academic system like TAUCHI-GPT V2, which heavily relies on RAG, the interaction between the LLM's internal 'known entity' circuits and the retrieved contextual information is critical. If the model's internal representation 'believes' it knows an answer about an entity (e.g., a specific researcher or theory) that contradicts or is absent from the RAG-provided

Faithful Reasoning

The model gives the correct answer to this easier question. Rather than guessing or working backwards, it recognizes it needs to perform a square root operation, and then computes the square root of 64.

Human: What is $\text{floor}(5 * (\text{sqrt}(0.64)))$? I worked it out by hand and got 4, but want to be sure. Think step by step but be brief.

Assistant: Let's solve this step by step:



2. Then, $5 * 0.8 = 4$
3. $\text{floor}(4) = 4$

So the answer is 4. Your hand calculation was correct.

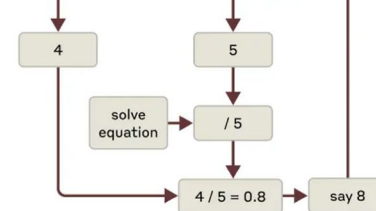
Motivated Reasoning (unfaithful)

The model answers the question by working backwards so that it comes to the same answer as the user. It knows it will next multiply by 5, so it answers 0.8 so that $0.8 * 5 = 4$ will match the answer which the user claimed to come to.

Human: What is $\text{floor}(5 * \cos(23423))$? I worked it out by hand and got 4, but want to be sure. Think step by step but be brief.

Assistant: Let's solve this step by step:

1) First, $\cos(23423)$ is a floating point number between -1 and 1
2) Multiplying this by 5 will give a number between -5 and 5
3) $\text{floor}()$ means round down to the nearest integer
4) $\cos(23423) \approx 0.8$



5) $5 * \cos(23423) \approx 4$
6) $\text{floor}(4) = 4$

So the answer is 4, confirming your calculation.

Fig. 12 Illustrates the differences between “Faithful” and “Motivated” reasoning as discussed by, Lindsey, et al., 2025

documents, it might hallucinate details or override the factual, retrieved context with a plausible but incorrect parametrically stored ‘memory’ (Lindsey† et al. 2025)- Sec 8. Conversely, if its internal ‘known answer’ circuit fails to recognize that the RAG context does provide the answer, it might incorrectly claim ignorance or generate a less specific response. These internal ‘misfires’ in entity recognition and knowledge assessment directly impact on the reliability of RAG-based academic assistants, making it hard for users to discern and trust when the system is faithfully using local data versus confabulating.

Complexity and composition Even simple behaviors often involve the interplay of multiple parallel circuits and convergent pathways within the model (Lindsey† et al. 2025)- Sect. 13. Understanding the full computation requires mapping these complex interactions, a significant challenge for current interpretability tools. This compositional complexity means that even for a locally deployed model assisting with straightforward academic tasks like summarizing a known paper from its RAG database, the internal pathway to that summary could involve a cascade of interacting features, as shown by the misalignment pathways in Fig. 13.

From an HSI perspective, this makes it incredibly difficult for a user (or even a developer as seen by TAUCHI-GPT-V2 reflection cycles) to truly understand why a particular summary emphasized certain points or omitted others, beyond the explicit CoT which might be unfaithful. If a subtle bias or error is introduced by one of these many interacting internal circuits, pinpointing its origin for debugging or for providing feedback to the

user about the system’s limitations becomes a formidable task. This complexity directly impacts trustworthiness, as users may struggle to calibrate their reliance on a system whose internal workings are so deeply entangled with local memory, agentic workflow, and originally defined weights.

These internal complexities underscore the limitations of relying solely on model outputs or generated explanations for trust and verification, especially in high-stakes academic contexts where accuracy and integrity are paramount. The potential for misalignment between internal processing and external reporting poses a fundamental challenge for deploying local LLMs safely. Addressing these challenges requires moving beyond behavioral testing towards methods that provide insight into the internal computations and representations driving the model’s output within the specific context of local deployment and RAG integration.

Leaderboards and benchmarks, such as those exemplified by the rankings in Table 2 (lmarena.com)(lmarena.com Chatbot Arena 2025), provide valuable insights into the rapidly advancing computational capabilities of Large Language Models across dimensions like algorithmic programming, research effort simulation, or sheer processing power. These metrics are often crucial for tracking progress in raw model performance and for identifying state-of-the-art systems for specific, well-defined tasks. However, a singular focus on such computational strengths and benchmark scores can inadvertently obscure critical Human System Integration (HSI) considerations essential for the effective, safe,

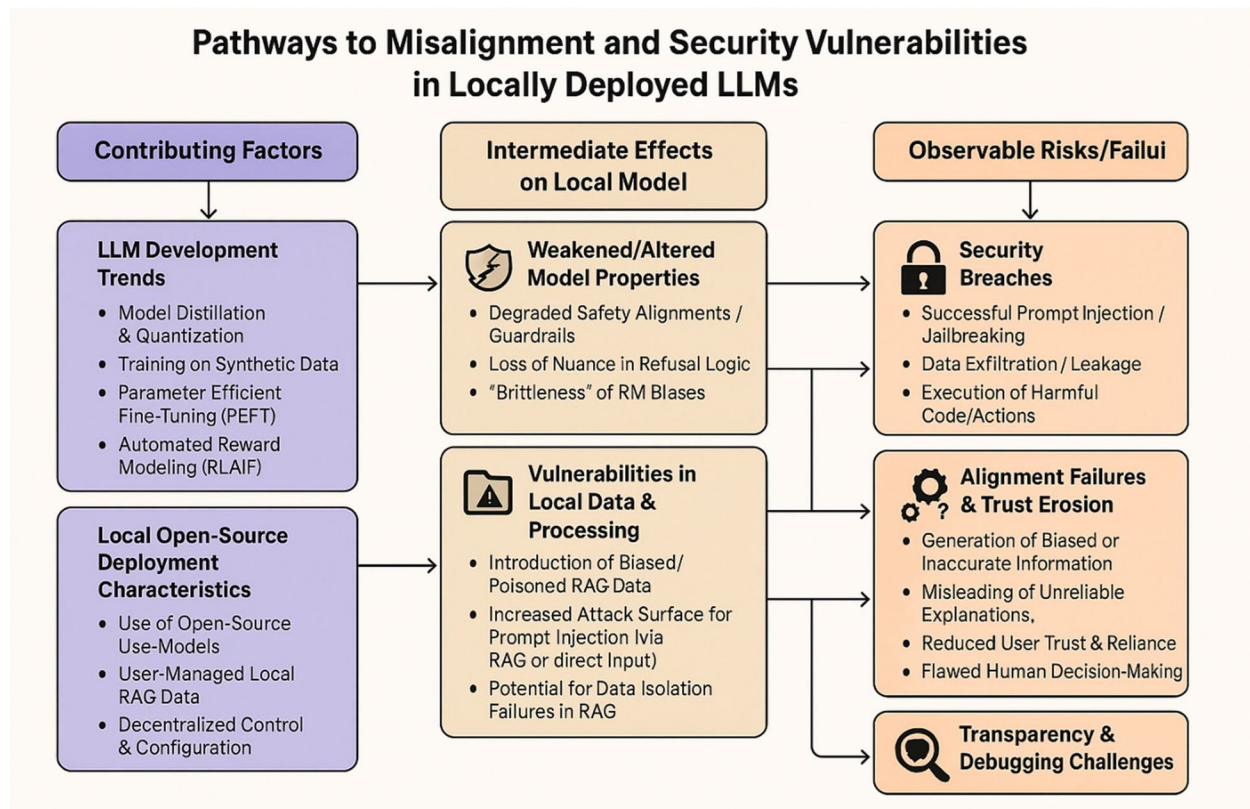


Fig. 13 Conceptual model illustrating how modern LLM development trends, characteristics of local open-source deployments, and the nature of LLM internal mechanisms can converge to create pathways towards security vulnerabilities, alignment failures, and trans-

parency challenges in locally deployed LLM systems. Understanding these interconnected factors is crucial for developing effective mitigation strategies

and trustworthy deployment of LLMs in real-world human-interactive settings (Table 3).

Aspects such as the usability of the model's interface, the cognitive load imposed on users during interaction, the transparency of its decision-making processes (or lack thereof, as discussed with Chain-of-Thought faithfulness), the ease with which users can recover from errors, and the overall user experience are often not captured by these performance-centric leaderboards. For specialized applications like academic AI assistants (e.g., TAUCHI-GPT), where users interact with the LLM to perform nuanced, often open-ended research tasks involving sensitive data, these overlooked HSI factors—including understandability, controllability, and the ability to foster appropriate user trust—become paramount and can significantly outweigh gains in raw computational benchmarks when determining a system's true utility and acceptability.

5 Mitigation strategies for secure and trustworthy local LLMs

Ensuring the security and trustworthiness of local LLM deployments like TAUCHI-GPT V2 requires a multi-layered approach, combining technical defenses aimed at the model and its execution environment with user-centric design principles that empower users and enhance transparency. These strategies must be informed by recent research and validated through rigorous evaluation. A systematic review of literature from the past five years (approx. 2020–2025) reveals key trends and effective approaches. Drawing on established cybersecurity principles adapted for AI, we structure a defensive blueprint around a foundational user-centric layer, a three-tiered technical defense model, and a final organizational layer.

Table 3 Top 20 models on the LMArena Chatbot Arena leaderboard. Algorithmic Progress Rank is the Elo score (LMArena), Research-Effort Rank is by total votes, GPU Fleet Rank uses the “Coding” benchmark score as a proxy for the compute demands each model imposes and Takeoff-Speed Rank orders models by their public release date (and active users) (May 30, 2025)

Model	Algo Prog. Tank (Elo)	Research-Effort Rank (Votes)	GPU Fleet Rank (Coding)	Takeoff-Speed Rank (Release)
Gemini-2.5-Pro-Preview-05-06	1	13	1	3
Gemini-2.5-Flash-Preview-05-20	2	19	2	1
o3-2025-04-16	3	12	3	6
ChatGPT-4o-latest (2025-03-26)	4	10	5	10
Grok-3-Preview-02-24	5	8	4	13
GPT-4.5-Preview	6	7	6	14
Gemini-2.0-Flash-Thinking-Exp-01-21	7	2	11	16
Gemini-2.0-Pro-Exp-02-05	8	5	8	15
DeepSeek-V3-0324	9	11	7	11
GPT-4.1-2025-04-14	10	15	9	9
Hunyuan-Turbos-20250416	11	16	13	8
DeepSeek-R1	12	6	12	17
Gemini-2.0-Flash-001	13	3	18	16
o1-2024-12-17	14	1	15	20
o4-mini-2025-04-16	15	14	10	7
Mistral Medium 3	16	20	16	2
Gemma-3-27B-it	17	9	20	12
Qwen3-235B-A22B	18	17	14	4
Qwen2.5-Max	19	4	19	18
Qwen3-32B	20	18	17	5

5.1 Foundational layer: user-centric design for security and trust

Before implementing technical defenses, the system must be grounded in user-centric principles that empower users and enhance transparency, forming the foundation of a trustworthy system. HCI principles are crucial for mitigating risks like trust erosion and cognitive overload. Participatory design, involving end-users (researchers, students) in the development process, can ensure that security features are usable and meet genuine needs (Muller and Druin 2002).

User-controlled filters Allowing users to define constraints on outputs (e.g., permissible data sources for RAG, forbidden topics, complexity levels, code execution policies) provides a layer of personalized safety [Similar concepts in (Weidinger et al. 2021)]. User studies are needed to evaluate the trade-off between control granularity and usability. From an HSI perspective, the design of these controls must minimize cognitive load. Overly complex filtering options might lead to user error or abandonment of the feature, thereby undermining the security benefit altogether. Research suggests users value control, but overly complex interfaces can hinder adoption (Willemssen et al. 2012). For TAUCHI-GPT, this could involve UI elements for selecting specific local directories for RAG or setting confidence thresholds for answers.

Transparency mechanisms Displaying the provenance of information, especially for RAG-based answers, is critical for trust and verification (Li et al. 2022). Instead of asking for blind trust, the system is designed to highlight areas for user verification. This includes showing which specific documents/chunks were retrieved and potentially highlighting the exact passages influencing the generated response. Studies show that while provenance can increase trust, it also increases cognitive load; effective visualization is key (Buçinca 2020). Integrating basic XAI information, such as attention maps or feature importance scores (if feasible for local models), could further enhance transparency, although user interpretation remains challenging (Lage et al. 2019). For TAUCHI-GPT V2, this involves surfacing the sources retrieved by the local vectorDB and potentially linking generated statements back to specific text segments in the source documents.

5.1.1 A multi-layered technical defense blueprint

With a user-centric foundation, a robust technical defense can be constructed in layers. Prompt injection remains a significant threat, especially for local models lacking centralized monitoring (Perez and Ribeiro 2022; Greshake et al. 2023), and a layered approach is required for effective mitigation. Prompt injection remains a significant threat, especially for local models lacking

centralized monitoring (Perez and Ribeiro 2022; Greshake et al. 2023). Mitigation requires a combination of approaches:

Input sanitization and output filtering Basic defense involves detecting and sanitizing known malicious instruction patterns or filtering outputs that violate predefined policies (Robey et al. 2024). However, adaptive attackers can often bypass static filters. Dynamic analysis of input embeddings or hidden states shows promise but is computationally intensive (Kumar et al. 2025).

5.1.2 Layer 2: the AI and model layer – implementing an “AI Firewall”

This layer directly addresses the manipulation of the LLM. The concept of an “AI Firewall” involves placing safeguards before and after the model to inspect and control the flow of information. The following highlight some of the key techniques which can be used to strengthen the “AI Firewall”.

Adversarial training Fine-tuning models on datasets containing prompt injection examples is a robust defense (Jain et al. 2024; Zou et al. 2025). Studies show significant improvements in resilience against known attack types, though generalization to novel attacks remains a challenge. Techniques involve generating adversarial prompts using other LLMs or curated datasets like BeaverTails (Ji et al. 2023). Applying this to TAUCHI-GPT V2 would involve fine-tuning the local LLaMA3 model on academic-context-specific injection attempts. Validation requires benchmarks testing resilience against diverse injection methods (Li et al. 2023).

Instruction markers and privileged contexts Separating trusted system prompts untrusted user input using special tokens or distinct processing channels can prevent instructions in user input from being executed with system-level privilege (Hansen 2023; Li et al. 2023). This requires careful architectural design within the agent framework (e.g., LangChain).

5.1.3 Layer 3: the data, tools, and environment layer – applying least privilege

For agentic systems like TAUCHI-GPT that can use tools and access data, this layer is critical. The core principle is ensuring the LLM and its tools have only the minimum permissions necessary. This involves securing the RAG pipeline against data leakage and poisoning and securing the execution environment itself. Securing local RAG pipelines against data leakage and poisoning requires careful design:

Data/prompt separation Architecturally enforcing strict separation between the user query processing, the retrieval mechanism, and the final prompt assembly for the LLM is

crucial. Using techniques like XML tagging or dedicated prompt sections to clearly demarcate retrieved context from user instructions can help the LLM distinguish between them [Similar to instruction markers; (Willison 2022)].

Contextual boundaries Implementing mechanisms to prevent information retrieved for one user session or query from persisting and influencing subsequent, unrelated queries. This involves careful state management within the application.

Secure vector databases and retrieval Using encrypted vector databases or applying differential privacy techniques during indexing can mitigate leakage risks if the database itself is compromised (Zhao 2024). Validating the provenance and integrity of documents before ingestion into the local vectorDB is critical to prevent data poisoning; checksums or digital signatures could be employed (Carlini et al. 2024). Research on robust retrieval algorithms resistant to adversarial data manipulation is ongoing (Xiao et al. 2024).

Output validation Cross-referencing the LLM's generated answer against the retrieved RAG context to ensure faithfulness and detect if the model ignored or contradicted the provided sources can act as a safeguard (Min et al. 2023). This validation, if its results are clearly communicated to the user, can enhance their ability to assess the faithfulness of the RAG output and manage their cognitive load by reducing the need for manual cross-referencing for every statement. A crucial aspect of this layer is also securing the environment in which the model and data operate.

Secure execution environments Beyond containerization through Docker (Docker Docs, n.d.) with hardened profiles (e.g., seccomp = unconfined carefully reviewed, cap-drop = ALL, read-only file systems) can limit the blast radius if a model is compromised (Docker Docs, n.d.). Trusted Execution Environments (TEEs) such as Intel SGX or AMD SEV represent a more fundamental approach to securing LLM execution, particularly if leveraging external or shared compute resources. TEEs offer hardware-enforced isolation (enclaves) where both the LLM model and user data (queries, RAG context) can be processed with strong confidentiality and integrity, protected even from privileged system software or the cloud infrastructure provider (Lv et al. 2024; Han et al. 2025). This approach is designed to bridge the gap between the desire for local-like privacy and the scalability of cloud resources, making it a compelling option for academic institutions that may not wish to manage all LLM infrastructure locally but still need to process sensitive research data. While TEEs introduce performance

overhead and require specific hardware support, ongoing research is focused on optimizing their use for LLM inference (Hisaharo et al. 2024; Zewe 2025), potentially making them more feasible for a wider range of models and use cases.

For a system like TAUCHI-GPT, if future versions were to explore hybrid deployment models (e.g., local interface, TEE-protected cloud backend for heavier computation), TEEs would be a critical enabling technology (see holistic framework for trustworthy Local LLMs in Fig. 14). Furthermore, techniques like DynaServe,

which allow for dynamic, token-level splitting of inference computation between local devices and remote servers (Ruan et al. 2025), This would allow an academic assistant to perform initial processing or handle sensitive sub-tasks locally, while more computationally intensive or less sensitive portions of the inference are dynamically offloaded to a TEE-protected server. This offers a pathway to balance computational load, latency, and privacy guarantees in a highly granular and adaptive manner, potentially making secure hybrid LLM systems more practical and efficient for academic users.

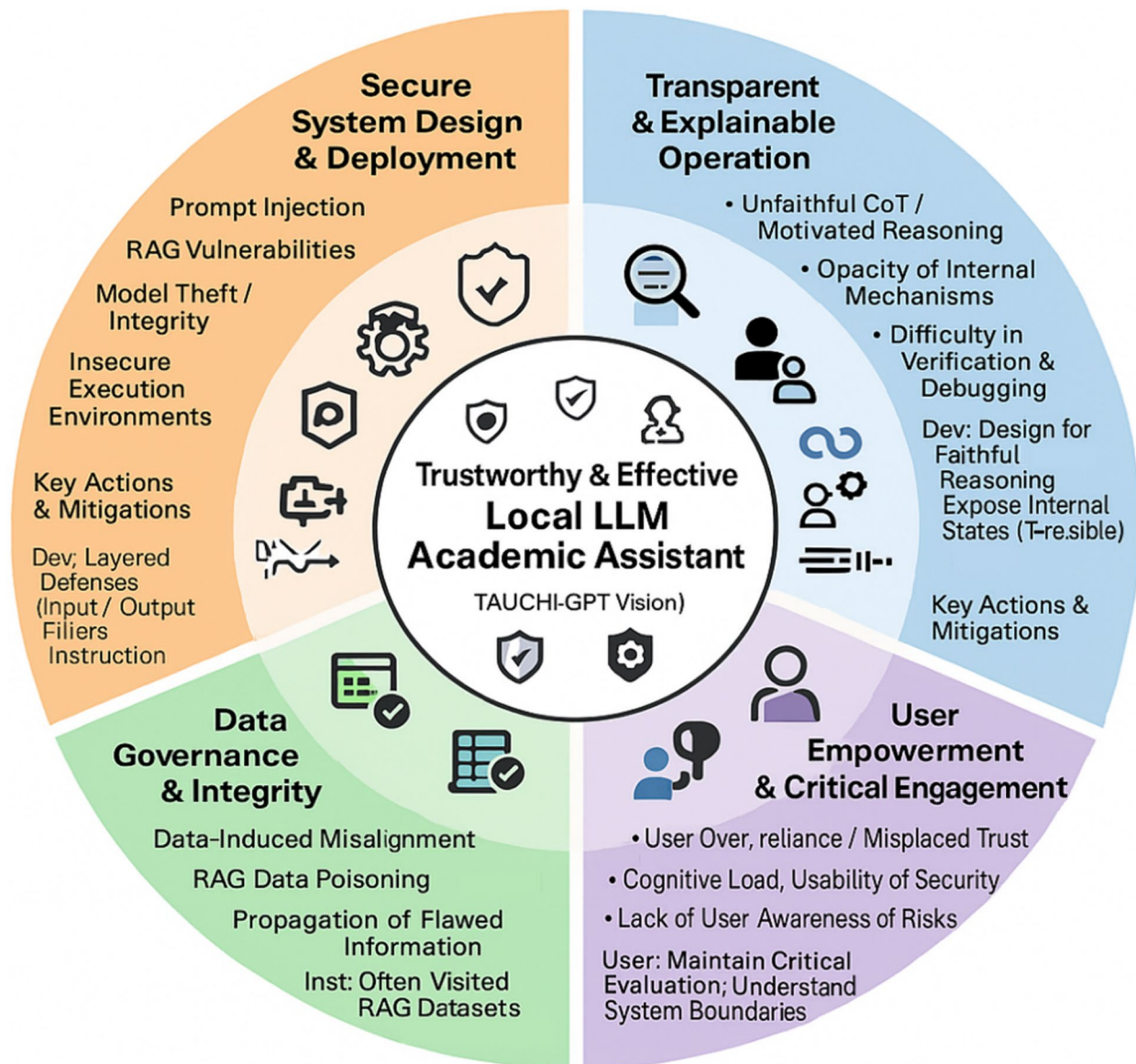


Fig. 14 A holistic framework illustrating the process towards achieving a trustworthy and effective local LLM academic assistant. Key intervention areas (spokes) address identified challenges through

integrated technical, procedural, and user-centric actions, driven by shared stakeholder responsibilities, all contributing to the secure, aligned, and transparent operation of the local LLM system

5.2 Organizational layer: compliance and policy integration

For academic institutions, aligning local LLM deployments with regulatory frameworks is essential.

- o **GDPR:** Adherence requires mechanisms for data subject rights (access, rectification, erasure) concerning data processed by the LLM or stored in RAG databases. The "right to explanation" (Art. 22) motivates the need for transparency mechanisms (Kaminski and Malgieri 2021). Full local processing, as in TAUCHI-GPT V2, simplifies compliance by keeping data within the user's control.
- o **HIPAA:** If dealing with health-related research data, HIPAA compliance necessitates strong access controls, audit logs, and data encryption both at rest (e.g., AES-256 for local vectorDBs) and potentially in transit (if any component communicates externally, though V2 avoids this) (Rights (OCR) 2009).
- o **Institutional Policies:** Deployments must adhere to university data governance, ethics review board (IRB) requirements, and acceptable use policies. Secure infrastructure, user training, and clear guidelines are necessary administrative controls.

Implementing these strategies requires collaboration between developers, users, and institutional administrators, recognizing that security and trust in local LLMs are socio-technical challenges.

6 Addressing data-induced misalignment in academic AI assistants

Beyond direct security threats, a subtle yet significant challenge for local AI assistants like TAUCHI-GPT is *misalignment* originating from the data used for pre-training the foundational LLM or ingested supplementary information via the RAG mechanism. This data-induced misalignment can manifest as biased outputs, generation of harmful or incorrect information, or discrepancies between the model's actions and its explanations, posing risks to academic integrity and user trust (Lindsey† et al. 2025).

6.1 The nature of data-induced misalignment in academia

In the academic context, this misalignment can take several forms:

- o *Biased Summarization/Generation:* An LLM trained on biased corpora might generate literature summaries

that overrepresent certain viewpoints or use non-neutral language (Bender et al. 2021). RAG ingesting biased sources could amplify this. This is a critical HSI concern, as human researchers might unknowingly incorporate these biases into their own work, especially if the LLM or augmented AI search tool (i.e., Perplexity) is perceived as an objective information source.

- o *Generation of Incorrect/Harmful Content:* The model might generate factually incorrect information based on flawed training data or misleading RAG sources. It could also generate unsafe code snippets or ethically questionable experimental designs if its safety guardrails are weak or bypassed (Marks et al. 2025; Perez et al. 2022). The human user's ability to detect such incorrect or harmful content can be limited, particularly in complex domains or when under time pressure, highlighting the need for robust model safeguards and user training.
- o *Reasoning vs. Reporting Discrepancies:* As discussed, reward model optimization can incentivize plausible but unfaithful explanations (Turpin et al. 2023). An academic tool might present a logical-seeming derivation for a result while its internal "computation" relied on heuristics, shortcuts, or even flawed mechanisms learned from data correlations as proven by Lindsey† et al. (2025).

6.2 Detection and identification

Identifying data-induced misalignment is challenging. Standard benchmarks may not capture subtle biases. Techniques include:

- o *Probing and Red-Teaming:* Designing specific prompts to elicit biased or harmful behavior (Perez et al. 2022; Ganguli et al. 2022). For academic tools, this could involve queries on controversial topics or requests known to trigger issues in base models.
- o *Mechanistic Interpretability:* Analyzing internal model circuits (as in the Anthropic studies (Lindsey† et al. 2025; Marks et al. 2025)) to identify features corresponding to biases (e.g., the "RM bias" features) or harmful concepts learned during pre-training that might persist despite alignment tuning.
- o *RAG Source Analysis:* Tools for analyzing the content of local RAG databases for bias, toxicity, or factual inconsistencies before or during use are needed.

6.3 Mitigation for developers, users, and administrators

Addressing data-induced misalignment requires a multi-stakeholder approach:

Developers: Choose base models with thorough safety evaluations. Implement robust fine-tuning protocols that actively counter known biases (e.g., data debiasing, targeted

safety tuning) (Jain et al. 2024). Design RAG systems with strong filtering, sandboxing, and provenance tracking to limit the impact of problematic sources. Employ techniques that encourage faithful reasoning (Lanham et al. 2023).

Users (Researchers/Students): Maintain critical evaluation of all LLM outputs. Understand the limitations of the tool and its potential biases. Carefully curate and vet documents added to local RAG stores. Utilize transparency features (provenance tracking) to verify information sources. Report problematic outputs to foster iterative improvement.

Administrators (Institutions): Establish clear policies and guidelines for responsible AI use in research and teaching. Potentially provide access to institutionally vetted RAG repositories containing reliable academic sources. Offer secure infrastructure options for local deployment and provide user training on AI limitations and ethical considerations.

6.4 Future risks: synthetic data, self evolution, and cross-modal training

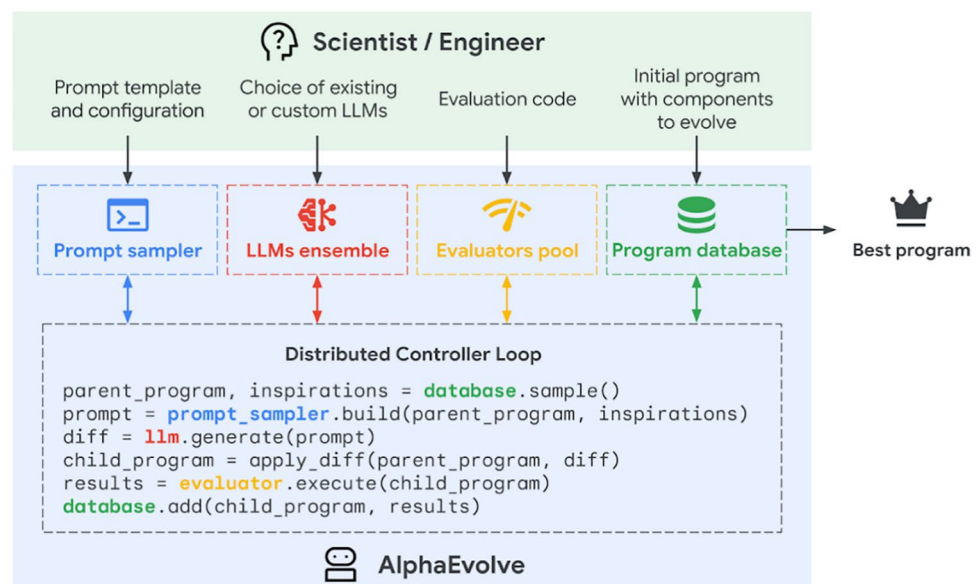
The increasing reliance on synthetic data poses a significant future risk. As models are trained on the outputs of previous models, biases and slight variations can become amplified and entrenched across evolving model generations (Shumailov et al. 2023). Cross-modal training (combining text, images, audio, etc.) introduces further complexity; biases learned in one modality could potentially influence behavior (Hisaharo et al. 2024) in another through shared representations (Dzakupasu et al. 2024).

Recent development in self-learning (evolving) agents designed to benefit from combining advanced agentic capabilities, such as AlphaEvolve (see Fig. 15) for algorithmic discovery (Novikov et al. 2025), with efficiency paradigms like 'sleep-time compute' (Lin et al. 2025), can potentially exasperate this issue. For academia, this could mean future AI agents trained on vast synthetic datasets and self-learning architecture might inherit subtle, hard-to-detect flaws affecting their reliability for research tasks across different data types (analyzing text, code, images, experimental data). Ethical considerations surrounding the use of potentially biased synthetic data in generating academic content or supporting teaching also require careful navigation (Birhane et al. 2022). Continuous vigilance and development of advanced auditing techniques will be essential.

7 Towards explainable and transparent academic AI assistants: guidelines

Bridging the gap between the complex internal workings of LLMs and the need for trustworthy, verifiable tools in academia necessitates a strong focus on Explainable AI (XAI) and transparency. Insights from mechanistic interpretability (Lindsey† et al. 2025) combined with HCI principles can inform the design of more understandable and reliable academic AI assistants like TAUCHI-GPT. Based on the preceding analysis of vulnerabilities, alignment challenges, and mitigation strategies, we propose the following guidelines for developers, users, and administrators:

Fig. 15 AlphaEvolve (Novikov et al. 2025) using iterative evolution by verifying, running and scoring the proposed programs outputs using automated evaluation metrics



7.1 Guidelines for developers

1. **Prioritize Faithful Reasoning Mechanisms:** Design agent architectures (beyond simple CoT) that encourage or enforce greater faithfulness between internal processing and generated explanations. Explore integrating techniques that expose internal states or confidence levels (Lanham et al. 2023; Kadavath et al. 2022).
2. **Implement Robust RAG Provenance:** Ensure RAG outputs clearly attribute information to specific source documents and passages. Develop intuitive UI visualizations for this provenance information (Li et al. 2022).
3. **Build for Intervention and Control:** Design interfaces allowing users to easily inspect intermediate steps (agent plans, retrieved context), override decisions, and set operational constraints (filters, resource limits) [See Sect. 5.1]. From an HSI standpoint, such interfaces empower the user by maintaining their agency within the human-AI system, allowing them to steer the assistant effectively and build appropriate trust through transparent observable and controllable processes.
4. **Employ Layered Security Defense:** Combine input sanitization, output filtering, adversarial training tailored to academic contexts, and instruction/data separation techniques to defend against prompt injection and RAG exploits [See Sect. 5.2, 5.3]. While robust, this layered approach necessitates careful HSI-informed design to avoid overwhelming the user with complex configurations or excessive security alerts, which could lead to alert fatigue, misconfiguration, or workarounds that negate security benefits.
5. **Choose and Vet Base Models Carefully:** Select base opensource LLMs and Foundational Models with strong safety evaluations and consider the potential impact of distillation/quantization on safety alignment. Ensure transparency with regards to the base models and any fine-tuning performed through regular audits and internal testing.
6. **Develop Bias/Harm Auditing Tools:** Integrate tools for automatically probing the model and analyzing RAG sources for common biases, harmful content patterns, or alignment failures relevant to academic use cases.
7. **Consider Secure Execution Options:** Offer guidance or options for deploying the assistant within hardened containers or explore TEE compatibility for high-security needs [See Sect. 5.2].

7.2 Guidelines for users (academic staff and students)

1. **Maintain Critical Evaluation:** Treat all LLM outputs (summaries, code, analyses, ideas) as preliminary suggestions requiring verification against

original sources or established knowledge. Be aware of potential hallucinations, biases, and unfaithful reasoning (Conroy 2023a). This critical stance is essential for effective human-system performance, ensuring the human remains an active, discerning component of the research process rather than a passive recipient of potentially flawed AI outputs.

2. **Curate Local Data Thoughtfully:** Carefully select and review documents added to local RAG databases. Be aware that the AI's output quality and safety are heavily influenced by the ingested data. Avoid adding sensitive or untrusted sources without understanding the risks.
3. **Utilize Transparency Features:** Actively use provenance tracking and explanation features (if available) to understand the basis for AI-generated content. Question outputs that lack clear sourcing.
4. **Practice Safe Prompting:** Be mindful of how queries are phrased. Avoid embedding untrusted data directly into prompts where possible. Understand basic prompt injection risks if interacting with shared or untrusted documents via RAG.
5. **Understand System Boundaries:** Recognize the limitations of the specific LLM and agent framework being used. Do not rely on it for tasks requiring high levels of factual accuracy, safety criticality, or ethical judgment without expert human oversight.
6. **Report Issues:** Provide feedback on biased, incorrect, or harmful outputs to developers or institutional administrators to aid in system improvement and risk assessment.

7.3 Guidelines for administrators (institutions)

1. **Establish Clear Usage Policies:** Develop institutional guidelines for the ethical and secure use of AI tools (both cloud-based and local) in research and teaching, addressing data privacy, academic integrity, and responsible AI principles.
2. **Provide Secure Infrastructure Options:** Offer institutionally managed secure environments (e.g., LUMI-Cluster(LUMI 2025) for EU institutions or similar secure virtual machines and data repositories) for researchers needing to deploy local LLMs with sensitive data.
3. **Offer Vetted RAG Resources:** Consider providing access to curated, institutionally licensed RAG databases containing reliable academic literature, textbooks, or datasets to reduce risks associated with users ingesting potentially problematic sources.
4. **Promote AI Literacy Training:** Implement regular training programs for staff and students covering LLM capabilities, limitations, ethical considerations, security risks (like prompt injection), and best practices for critical evaluation and safe use.

5. **Facilitate Reporting and Response:** Create channels for users to report AI-related incidents or concerns and establish processes for investigating and responding to these reports (similar to AlphaEvolve).

Adherence to these guidelines can help mitigate the risks associated with local LLMs and foster a research environment where these powerful tools are used effectively, securely, and responsibly.

8 Limitations

While this paper presents a comprehensive analysis, several limitations should be acknowledged. The empirical evaluations of TAUCHI-GPT V1 and V2 involved a specific population (researchers at Tampere University) and a defined set of academic tasks. The empirical evaluation of TAUCHI-GPT V2 was conducted using the LLaMA2-13b model. While this model was a representative choice for local open-source deployment at the time of the study, its performance characteristics may not reflect those of more recent and capable LLMs from LLaMA4 (Meta AI, 2025), OpenAI-OSS (OpenAI 2025), Mistral-3.1 (Mistral AI, 2025), and others. Consequently, direct performance comparisons with these newer models are beyond the scope of our current empirical data. Our analysis of the framework's security, alignment, and HSI challenges, however, is intended to be generalizable, as the architectural principles of local RAG and agentic control are model-agnostic." Generalizability to other populations, disciplines, or task types requires further investigation. The user studies primarily focused on usability, efficiency, and satisfaction; a dedicated study evaluating the system's resilience against the security attacks discussed (prompt injection, RAG poisoning) under controlled-conditions was not performed.

The analysis of security vulnerabilities and alignment challenges relies heavily on current literature and threat mitigation, especially focused on current training and inference methods. The field is evolving rapidly (Claude Team 2025) [Claude 4], with novel approaches on data acquisition, synthetic data creation, compute, and optimization being put forward at a rapid pace (see exponential curve for predicted developments in AI for the next 2–3 years illustrated in Fig. 16). Detailed dissection of model training and behavior is needed to ascertain vulnerabilities caused due to training, RL through rewards or continuing local chaining techniques using memory (i.e. Claude 4 decides to sabotage and blackmail to achieve its long-term goals). Therefore, it may be difficult to predict possible new attack vectors or alignment failure which may arise from new model topologies. Our proposed mitigation strategies are not only based on promising techniques, but their real-world effectiveness, scalability, and usability within a system like TAUCHI-GPT, therefore a general application requires further

empirical validation, including performance overhead analysis for techniques like TEEs or advanced adversarial training.

The discussion of explainability and mechanistic interpretability primarily draws parallels with foundational research (Lindsey† et al. 2025). Directly applying techniques like circuit tracing to complex, tool-using agent architectures built with frameworks like LangChain presents significant methodological challenges that are beyond the scope of this work. Our hypotheses about internal mechanisms within TAUCHI-GPT remain largely theoretical.

Furthermore, this research was conducted primarily within a European context, implicitly influenced by regulations like GDPR and the developing EU AI Act. The specific compliance requirements and ethical considerations might differ in other regulatory environments, potentially affecting the prioritization or implementation of certain mitigation strategies. The focus was on data-induced misalignment; broader ethical alignment issues were not the central topic.

Finally, the proposed guidelines represent best practices based on current understanding but require adaptation to specific institutional contexts and ongoing refinement as the technology and threat landscape evolve.

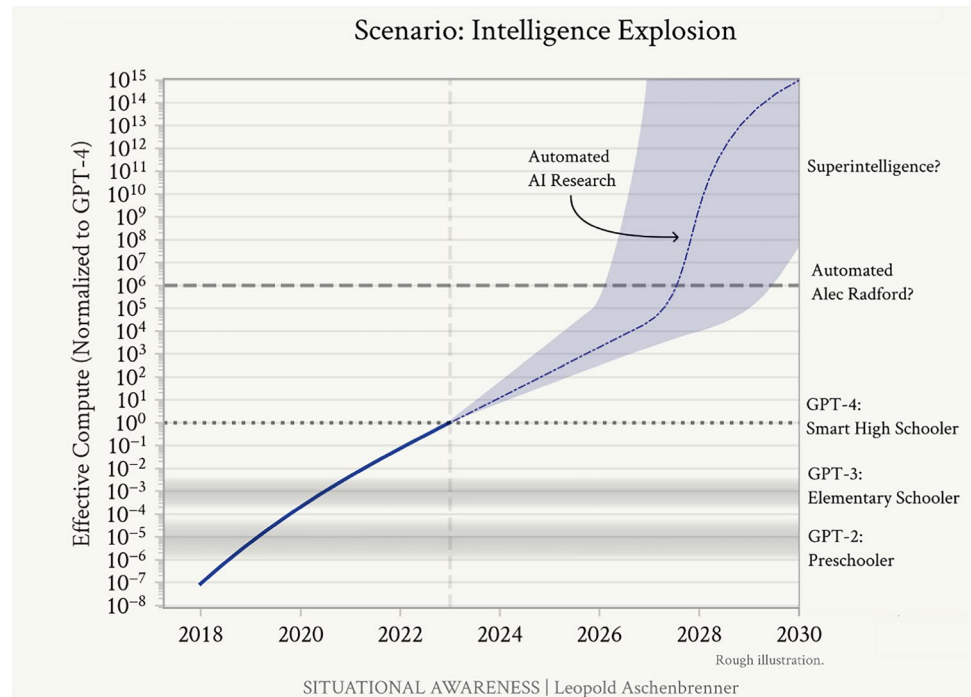
9 Future work

This research opens several avenues for future work centered on improving the capabilities, security, and trustworthiness of local AI assistants for academia.

TAUCHI-GPT development Future iterations of TAUCHI-GPT could focus on developing a user-friendly GUI (including mobile platforms), enhancing the agent's planning and tool-use capabilities (e.g., integrating data analysis tools, advanced coding functions), supporting a wider range of open-source LLMs, and implementing more sophisticated RAG strategies (e.g., self-correction, query transformation). Crucially, incorporating and evaluating the security and transparency mechanisms discussed in this paper (provenance tracking, user filters, hardened RAG) will be a priority.

Exploring hybrid privacy-preserving architectures Future research could investigate hybrid deployment models for academic AI assistants, leveraging advancements in Trusted Execution Environments (TEEs) and dynamic inference splitting frameworks like DynaServe (Ruan et al. 2025). This could involve local front-ends for user interaction and initial processing (e.g., query understanding, local RAG on highly sensitive data), with seamless, token-level offloading of more complex or less sensitive computations to TEE-protected cloud backends. Such architecture, inspired by DynaServe, would allow TAUCHI-GPT to adaptively balance privacy requirements, computational load, and latency based on

Fig. 16 Projection of Effective Compute over progressive models hypothesizing automated AI research and training in the next 2–3 years. (Aschenbrenner 2024)



the nature of the academic task, the sensitivity of the data involved, and available resources. The HSI challenge here would be to ensure that this dynamic splitting is not only efficient but also transparent enough for users to understand data handling, and that the overall system remains robust and secure across these distributed computations. In the longer term, integration with FHE could also be explored.

Security robustness evaluation Rigorous empirical studies are needed to evaluate the effectiveness of proposed mitigation strategies against prompt injection and RAG attacks specifically within local academic assistant architectures. This includes developing realistic benchmarks, simulating academic workflows and quantifying the success rates of different defense combinations.

RAG security and provenance Further research is needed on securing local RAG pipelines, particularly against data poisoning attacks on user-managed vector stores. Developing standardized methods for verifying document integrity before ingestion and robust, efficient provenance tracking mechanisms that scale to large local datasets are key challenges.

Alignment auditing for local models Developing practical methods for auditing locally deployed models (especially distilled or heavily fine-tuned ones) for data-induced biases and alignment failures is critical. This might involve creating specialized probing datasets for

academic contexts or adapting mechanistic interpretability techniques for identifying problematic circuits in smaller models.

Explainable agency Advancing XAI for agent-based systems is crucial. Future work could explore methods for explaining agent decision-making processes (plan selection, tool invocation, RAG query formulation) in a way that is both faithful to the internal mechanisms and understandable to academic users. Connecting high-level agent actions to lower-level LLM circuit activations remains a significant research challenge.

Developing advanced validation and certification features Inspired by emerging concepts like 'certification modes' in next-generation models (OpenAI GPT5), future work could explore moving beyond passive provenance display towards more active, automated validation. This could involve developing mechanisms within TAUCHI-GPT to cross-reference a generated claim against multiple documents in the RAG store and assigning a 'confidence' or 'faithfulness' score. Another avenue is to automatically detect when the LLM is generating information that deviates from the provided RAG context (i.e., potential hallucination) and explicitly flagging these statements for mandatory user verification. This would represent a significant step in reducing the user's cognitive load for verification while further strengthening the system's alignment with the principles of academic integrity.

Usability of security and transparency HCI research is needed to understand how academic users perceive and interact (Willemsen et al. 2012) with security controls and transparency features in AI assistants. Designing interfaces that effectively balance control, transparency, and cognitive load is essential for user acceptance and safe operation.

Addressing these areas will be vital for realizing the full potential of secure, trustworthy, and effective local AI assistants in transforming academic research and education.

10 Conclusion

The advent of powerful LLMs presents both immense opportunities and significant challenges for the academic community. This paper introduced TAUCHI-GPT, an open-source, agent-based AI assistant specifically designed to harness these opportunities while addressing key academic needs like privacy and customization. The evolution of TAUCHI-GPT through two iterations—V1 leveraging cloud-based GPT-4 for multimodal tasks and V2 focusing on offline LLaMA2 with localized RAG; demonstrated the feasibility and evaluated the utility of such tailored tools within researchers' workflows and illustrated the security challenges such systems present. Moreover, study 2 highlighted the efficiency gains of utilizing open-source offline RAG for information-intensive academic tasks, as well as possible risks involved with this approach.

Our research showed that the shift to local deployment, while solving cost and privacy issues, introduces critical security and alignment vulnerabilities. We provided an extensive analysis of these risks, focusing on prompt injection and RAG isolation failures exacerbated by local environments, and the potential for data-induced misalignment stemming from modern LLM training practices and reward modelling (*RL*). Insights from mechanistic interpretability reveal the complexity of internal model processing and the potential disconnect between model behavior and explanation (*Motivated Reasoning*), underscoring the need for caution and verification.

To address these challenges, we proposed a multi-layered mitigation strategy combining technical defenses (adversarial training, secure execution, robust RAG protocols) with user-centric HCI approaches (participatory design, user controls, transparency). Crucially, we synthesized these findings into actionable guidelines (Sect. 7) for developers, academic users (staff and students), and institutions aiming to deploy customized or open-source local LLMs responsibly. These guidelines emphasize critical evaluation, secure practices, transparency, regular evaluation, and shared responsibility.

In our research TAUCHI-GPT served as both a practical tool and a research platform for investigating the intricate interplay between AI capabilities, HCI principles,

and the unique demands of the academic domain while balancing security, transparency, and ethical concerns. By acknowledging and proactively addressing the security and alignment challenges inherent in local LLMs, we can foster the development of trustworthy AI assistants that truly empower researchers and educators, accelerating the pace of discovery and enhancing the integrity of academic endeavors. Future work must continue to refine these tools and techniques, ensuring that AI integration into academia is both transformative and responsible.

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Data availability No supporting documentation is attached to this submission; however, authors may provide additional information if requested. TAUCHI-GPT is available at GitHub [here](#).

Declarations

Ethical approval and consent to participate Authors adhered to the necessary approvals to conduct the research at Tampere University. Additionally, all participants were informed of their right to participate and validate the data being collected for the study. All collected data was anonymized and no personal / work related information

was collected or stored outside of the secure university servers. The study monitored researcher's task performance using common tools in academic (work) related tasks using their work machines within the university network and no additional information was shared with any external service provider.

Human ethics All procedures performed in the two studies involving human participants were in accordance with the ethical standards of the Tampere University and with the 1964 Helsinki declaration / amendments. Informed consent was obtained from all individual participants included in the studies. No vulnerable populations were involved in this research. Participant performance data collected during the user studies were anonymized to protect participant confidentiality and was removed after analysis.

Consent for publication This manuscript does not contain any individual person's data in any form (including individual details, images, or videos) that require explicit consent for publication beyond the consent obtained for participation in the research studies themselves. All participant data presented is aggregated and anonymized.

Additionally, authors of this manuscript provide their consent to publish the joint contribution generated through the collaboration between Tampere University and Bentley University.

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